

Polynomials Trajectory Planner

Preliminary Documentation

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Abstract

Symbology, nomenclature definitions. Theory of the polynomial profiles.

Contents

1	Symbols And Nomenclature	5
1.1	Intro And Scope	5
1.2	Guiding Conventions	6
1.3	Time Variables	6
1.4	Position And Trajectory In 1D	6
1.5	Derivatives In 1D	7
1.6	Boundary Conditions In 1D	7
1.7	State Notation	7
1.8	Position And Trajectory In 3D	8
1.9	Derivatives In 3D	8
1.10	Boundary Conditions In 3D	8
1.11	Segments, Multi-Segment Trajectories, And Blending	9
1.12	Constraint Notation	9
1.13	Reserved Orientation Notation	10
1.14	Procedure For Later Documents	10
1.15	Worked Symbolic Considerations	10
1.16	Open Questions	11
1.17	Related Documents	11
2	Ninth-Degree Polynomial Theory	11
2.1	Intro And Scope	11
2.2	Symbols Used	11
2.3	Why Degree Nine Is A Natural First Case	12
2.4	Canonical Form In The Time Domain	12
2.4.1	Reading The Canonical Form	12
2.5	Horner Form In The Time Domain	12
2.5.1	How Horner Form Is Built	13
2.6	Canonical Form Vs Horner Form	13
2.6.1	Canonical Form: Advantages	13
2.6.2	Canonical Form: Limitations	13
2.6.3	Horner Form: Advantages	13
2.6.4	Horner Form: Limitations	14
2.6.5	Practical Interpretation	14
2.7	General Derivative Rule	14
2.8	Derivatives Of The Ninth-Degree Polynomial	14
2.8.1	First Derivative: Velocity	14

2.8.2	Second Derivative: Acceleration	14
2.8.3	Third Derivative: Jerk	15
2.8.4	Fourth Derivative: Snap	15
2.8.5	Fifth Derivative: Crackle	15
2.8.6	Sixth Derivative: Pop	15
2.9	What Changes In 3D	15
2.10	Why Fewer Derivatives May Be Enough	16
2.10.1	Position And Velocity Only	16
2.10.2	Up To Acceleration	16
2.10.3	Up To Jerk	16
2.11	Why More Derivatives Can Be Useful	16
2.11.1	First Reason: Endpoint Specification Richness	16
2.11.2	Second Reason: Higher-Order Smoothness	16
2.11.3	Third Reason: Analytical Support For Later Constraints	16
2.12	Are Crackle And Pop Physically Necessary	17
2.13	The Special Role Of Snap In Degree Nine	17
2.14	A Practical Evaluation Procedure	17
2.15	Worked Symbolic Considerations	17
2.15.1	Example 1: Full Ninth-Degree Segment In 1D	17
2.15.2	Example 2: Why Higher Derivatives Matter For Extrema	18
2.15.3	Example 3: 3D Interpretation	18
2.16	Open Questions	18
2.17	Related Documents	18
3	Boundary Conditions	18
3.1	Intro And Scope	18
3.2	Symbols Used	19
3.3	What A Boundary Condition Is	19
3.4	Why The Number Of Boundary Conditions Depends On Degree	19
3.5	The Natural Symmetric Endpoint Pattern	20
3.6	Three Condition Classes	20
3.6.1	Necessary Conditions	20
3.6.2	Possible Conditions	21
3.6.3	Redundant Conditions	21
3.7	Condition Catalogue In 1D	21
3.8	Degree 3	22
3.8.1	Polynomial Structure	22
3.8.2	Necessary Conditions	22
3.8.3	Possible Conditions	22
3.8.4	Redundant Conditions	22
3.9	Degree 5	22
3.9.1	Polynomial Structure	22
3.9.2	Necessary Conditions	23
3.9.3	Possible Conditions	23
3.9.4	Redundant Conditions	23
3.10	Degree 7	23
3.10.1	Polynomial Structure	23
3.10.2	Necessary Conditions	23
3.10.3	Possible Conditions	24
3.10.4	Redundant Conditions	24
3.11	Degree 9	24
3.11.1	Polynomial Structure	24
3.11.2	Necessary Conditions	24
3.11.3	Possible Conditions	24

3.11.4	Redundant Conditions	24
3.12	Summary Table For 1D	25
3.13	Independence Matters As Much As Count	25
3.14	What Changes In 3D	25
3.15	Relation To The Data Model	26
3.16	Worked Symbolic Considerations	26
3.16.1	Example 1: Cubic Segment	26
3.16.2	Example 2: Quintic Segment	27
3.16.3	Example 3: Ninth-Degree Segment	27
3.17	Open Questions	27
3.18	Related Documents	27
4	Polynomial Coefficient Determination	28
4.1	Intro And Scope	28
4.2	Symbols Used	28
4.3	Problem Statement	28
4.4	Core Idea	29
4.5	Absolute-Time Formulation	29
4.5.1	General Polynomial	29
4.5.2	General Derivative Rule	29
4.5.3	Derivative Row Operators	29
4.6	Degree-Specific Linear Systems In Absolute Time	30
4.6.1	Degree 3	30
4.6.2	Degree 5	31
4.6.3	Degree 7	31
4.6.4	Degree 9	31
4.7	Step-By-Step Procedure In Absolute Time	32
4.8	Why Normalized Time Is Worth Introducing	32
4.9	Derivative Scaling Between t And τ	33
4.10	Degree-9 System In Normalized Time	33
4.11	How The Lower Degrees Fit The Same Normalized Pattern	34
4.11.1	Degree 3	35
4.11.2	Degree 5	35
4.11.3	Degree 7	35
4.11.4	Degree 9	35
4.12	Absolute Time Vs Normalized Time	35
4.12.1	Absolute Time: Main Advantages	35
4.12.2	Absolute Time: Main Limitations	35
4.12.3	Normalized Time: Main Advantages	36
4.12.4	Normalized Time: Main Limitations	36
4.12.5	Practical Interpretation For This Project	36
4.13	Recommended Determination Strategy	36
4.14	Independence And Solvability	36
4.15	Relation To The Data Model	37
4.16	What Changes In 3D	37
4.17	Worked Symbolic Considerations	38
4.17.1	Example 1: Degree 3	38
4.17.2	Example 2: Degree 9 In Normalized Time	38
4.17.3	Example 3: Why Normalized Time Helps In Multi-Segment Work	38
4.18	Open Questions	38
4.19	Related Documents	39
5	Trajectory Constraints	39
5.1	Intro And Scope	39

5.2	Symbols Used	39
5.3	Constraint Context	40
5.4	Core Principle: Candidate Points For Extrema	40
5.5	Why Higher Derivatives Matter For Extrema	40
5.6	Position Extrema In 1D	40
5.6.1	General Rule	40
5.6.2	Degree-9 Specialization	41
5.6.3	Important Interpretation	41
5.7	Velocity Extrema In 1D	41
5.7.1	General Rule	41
5.7.2	Degree-9 Specialization	41
5.7.3	Absolute Velocity	42
5.8	Acceleration Extrema In 1D	42
5.8.1	General Rule	42
5.8.2	Degree-9 Specialization	42
5.8.3	Absolute Acceleration	42
5.9	Jerk Extrema In 1D	43
5.9.1	General Rule	43
5.9.2	Degree-9 Specialization	43
5.9.3	Absolute Jerk	43
5.10	Snap Extrema In 1D	43
5.10.1	General Rule	43
5.10.2	Degree-9 Specialization	43
5.11	Summary Of Stationarity Equations	44
5.12	Symbolic Setup Vs Numerical Root Extraction	44
5.13	Generic 1D Extrema Algorithm	44
5.14	Practical Root-Finding Note	45
5.15	Imposing A Velocity Limit In 1D	45
5.15.1	Problem Form	45
5.15.2	Verification Step	45
5.15.3	Constraint-Imposition Strategy	45
5.15.4	Duration Search	46
5.16	Imposing An Acceleration Limit In 1D	46
5.16.1	Problem Form	46
5.16.2	Verification Step	46
5.16.3	Constraint-Imposition Strategy	46
5.17	Imposing A Jerk Limit In 1D	47
5.17.1	Problem Form	47
5.17.2	Verification Step	47
5.17.3	Constraint-Imposition Strategy	47
5.18	A Caution About Time Scaling	47
5.18.1	Special Normalized-Profile Case	47
5.19	Position, Velocity, Acceleration, Jerk, And Snap In Normalized Time	48
5.20	What Changes In 3D	48
5.20.1	Component-Wise Analysis	48
5.20.2	Vector-Magnitude Analysis	48
5.21	Recommended 3D Interpretation For This Project	49
5.22	Relation To The Data Model	49
5.23	Generic Constraint-Verification Algorithm	50
5.24	Generic Constraint-Imposition Algorithm	50
5.25	Worked Symbolic Considerations	50
5.25.1	Example 1: Position Extrema	50
5.25.2	Example 2: Velocity Limit	51
5.25.3	Example 3: 3D Constraint Interpretation	51

5.26	Open Questions	51
5.27	Related Documents	51
6	Constant Velocity Segment	52
6.1	Intro And Scope	52
6.2	Symbols Used	52
6.3	Path Vs Trajectory: Why This Topic Is Special	52
6.4	First Key Fact: Exact Constant Velocity Is Very Restrictive	53
6.5	1D Case: Exact Constant Velocity Segment	53
6.6	Geometric Interpretation	53
6.7	Duration Of The Constant-Velocity Part	54
6.8	Why A Multi-Segment Reformulation Is Needed	54
6.9	Entry And Exit States In 1D	54
6.10	Baseline 1D Construction Strategy	55
6.11	How The Start And End Positions May Be Specified In 1D	55
6.11.1	Absolute Specification	55
6.11.2	Percentage Specification	55
6.12	1D Consistency Checks	55
6.13	A Simpler But Less Smooth 1D Alternative	56
6.14	3D Case: Two Different Meanings Of “Constant Velocity”	56
6.14.1	Constant Velocity Vector	56
6.14.2	Constant Speed Along The Original 3D Path	56
6.15	Decided 3D Interpretation For This Requirement	57
6.16	3D Path-Preserving Strategy: Arc-Length Reparameterization	57
6.17	Important Consequence In 3D	58
6.17.1	Implementation Note: Where Numerical Methods Are Actually Needed	58
6.18	3D Straight-Chord Alternative	59
6.19	How The Start And End Positions May Be Specified In 3D	59
6.19.1	Absolute Specification	59
6.19.2	Percentage Specification Along The Path	59
6.20	Transition Segments Around The Constant-Speed Part	60
6.20.1	1D Transition Targets	60
6.20.2	3D Transition Targets	60
6.21	Baseline Algorithm For Requirement 12	61
6.22	Relation To The Data Model	61
6.23	Worked Symbolic Considerations	61
6.23.1	Example 1: 1D Exact Constant-Velocity Middle Segment	61
6.23.2	Example 2: 3D Path-Preserving Constant-Speed Segment	62
6.23.3	Example 3: 3D Straight-Chord Alternative	62
6.24	Open Questions	62
6.25	Related Documents	63

1 Symbols And Nomenclature

1.1 Intro And Scope

This document defines the notation used throughout the theory library and, later, in the software implementation notes. Its purpose is to remove ambiguity before entering derivations, algorithms, and design decisions.

The project works with polynomial trajectories in 1D first and then extends the same concepts to 3D when needed. The 2D case is intentionally omitted because it is considered a direct conceptual subset of the 3D case.

Orientation is included here only as reserved notation. Full theoretical treatment of orientation is explicitly postponed to a later phase.

1.2 Guiding Conventions

- Time-dependent quantities are written as functions of time, for example $x(t)$.
- Scalar quantities are written in plain italic notation, for example $x(t)$, $v(t)$, T .
- Three-dimensional vectors are written in bold, for example $\mathbf{p}(t)$.
- Initial and final values use subscripts 0 and f , for example x_0 , x_f , $\mathbf{p}_0$, $\mathbf{p}_f$.
- Segment indices use subscripts k , for example $x_k(t)$.
- Normalized time is written as τ , while absolute time is written as t .

1.3 Time Variables

The following symbols are used for time:

- t : absolute time variable.
- t_0 : initial time.
- t_f : final time.
- $T = t_f - t_0$: motion duration.
- τ : normalized time variable, usually defined as

$$\tau = \frac{t - t_0}{T}, \quad \tau \in [0, 1].$$

This distinction is useful because some derivations are clearer in absolute time, while others are more compact in normalized time.

1.4 Position And Trajectory In 1D

In one dimension, the trajectory is represented by a scalar position law:

$$x(t)$$

For a polynomial of degree n , the canonical form is:

$$x(t) = \sum_{i=0}^n a_i t^i$$

where:

- a_i are the polynomial coefficients;
- n is the polynomial degree.

When the project specifically refers to the ninth-degree polynomial, the notation becomes:

$$x(t) = a_0 + a_1 t + a_2 t^2 + \cdots + a_9 t^9$$

The corresponding Horner form is written as:

$$x(t) = (((((((((a_9 t + a_8)t + a_7)t + a_6)t + a_5)t + a_4)t + a_3)t + a_2)t + a_1)t + a_0)$$

This document only fixes the notation. The analytical comparison between canonical and Horner form belongs to `NinthDegreePolynomialTheory`.

1.5 Derivatives In 1D

The project adopts the following derivative chain:

- position: $x(t)$
- velocity: $\dot{x}(t)$
- acceleration: $\ddot{x}(t)$
- jerk: $x^{(3)}(t)$
- snap: $x^{(4)}(t)$
- crackle: $x^{(5)}(t)$
- pop: $x^{(6)}(t)$

For quick reading, the shorthand notation below is also accepted when useful:

$$v(t) = \dot{x}(t), \quad a(t) = \ddot{x}(t), \quad j(t) = x^{(3)}(t), \quad s(t) = x^{(4)}(t)$$

where:

- $v(t)$ denotes velocity;
- $a(t)$ denotes acceleration;
- $j(t)$ denotes jerk;
- $s(t)$ denotes snap.

For higher derivatives beyond snap, the explicit derivative notation $x^{(5)}(t)$ and $x^{(6)}(t)$ is preferred because it is less ambiguous than introducing too many extra letters.

1.6 Boundary Conditions In 1D

A boundary condition is a value imposed at the start or end of a segment. The standard notation is:

$$x_0, x_f, v_0, v_f, a_0, a_f, \dot{j}_0, \dot{j}_f, s_0, s_f$$

These symbols mean:

- x_0, x_f : initial and final position;
- v_0, v_f : initial and final velocity;
- a_0, a_f : initial and final acceleration;
- $\dot{j}_0, \dot{j}_f$: initial and final jerk;
- s_0, s_f : initial and final snap.

Not every polynomial degree requires the same set of boundary conditions. In later documents, boundary conditions will always be classified as:

- necessary;
- possible;
- redundant.

This classification depends on the polynomial degree under study.

1.7 State Notation

When it is useful to describe the motion condition at a single endpoint, the project uses the concept of state.

In 1D, a state can be written as:

$$\mathcal{S}_{1D} = (x, v, a, j, s)$$

or, if endpoint notation is needed:

$$\mathcal{S}_0 = (x_0, v_0, a_0, j_0, s_0), \quad \mathcal{S}_f = (x_f, v_f, a_f, j_f, s_f)$$

The symbol $\mathcal{S}$ (calligraphic) is used here as a compact state container. It is deliberately distinct, both visually and semantically, from s / $\mathbf{s}$, which remain reserved exclusively for snap (see *Derivatives In 1D*) — no disambiguation is required when both appear in the same context, since the two symbols no longer collide.

1.8 Position And Trajectory In 3D

In three dimensions, position is represented by a vector:

$$\mathbf{p}(t) = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix}$$

The three scalar component laws are treated with the same notation already defined for 1D.

This means that a 3D polynomial trajectory can be understood as three coordinated scalar polynomial laws:

$$x(t), \quad y(t), \quad z(t)$$

or, compactly:

$$\mathbf{p}(t) = [x(t), y(t), z(t)]^T$$

1.9 Derivatives In 3D

The derivative hierarchy extends component-wise:

$$\dot{\mathbf{p}}(t), \quad \ddot{\mathbf{p}}(t), \quad \mathbf{p}^{(3)}(t), \quad \mathbf{p}^{(4)}(t)$$

The compact names are:

- velocity vector: $\mathbf{v}(t) = \dot{\mathbf{p}}(t)$
- acceleration vector: $\mathbf{a}(t) = \ddot{\mathbf{p}}(t)$
- jerk vector: $\mathbf{j}(t) = \mathbf{p}^{(3)}(t)$
- snap vector: $\mathbf{q}(t) = \mathbf{p}^{(4)}(t)$

The symbol $\mathbf{q}(t)$ is reserved here for vector snap only to avoid overloading the scalar letter s . If this creates confusion with quaternion notation in a future document, the explicit form $\mathbf{p}^{(4)}(t)$ should be preferred.

1.10 Boundary Conditions In 3D

The endpoint notation extends directly from the 1D case:

$$\mathbf{p}_0, \mathbf{p}_f, \mathbf{v}_0, \mathbf{v}_f, \mathbf{a}_0, \mathbf{a}_f, \mathbf{j}_0, \mathbf{j}_f, \mathbf{p}_0^{(4)}, \mathbf{p}_f^{(4)}$$

Each vector quantity contains three scalar components. For example:

$$\mathbf{v}_0 = \begin{bmatrix} v_{x,0} \\ v_{y,0} \\ v_{z,0} \end{bmatrix}$$

This reinforces the main project idea: 3D treatment is built from the 1D theory, then extended by vector composition.

1.11 Segments, Multi-Segment Trajectories, And Blending

For a single segment, the trajectory law may be written as:

$$x_k(t)$$

in 1D, or:

$$\mathbf{p}_k(t)$$

in 3D, where k is the segment index.

A complete multi-segment trajectory is represented as an ordered set:

$$\mathcal{T} = \{\text{segment}_1, \text{segment}_2, \dots, \text{segment}_N\}$$

When blending is introduced between consecutive segments, the project uses the following notation:

- $b_{k \rightarrow k+1}^{start}$: point where blending toward the next segment starts;
- $b_{k \rightarrow k+1}^{end}$: point where blending from the previous segment ends.

If the context is explicitly geometric, those symbols may be written as positions:

$$x_{b,k}^{start}, \quad x_{b,k}^{end}$$

in 1D, or:

$$\mathbf{p}_{b,k}^{start}, \quad \mathbf{p}_{b,k}^{end}$$

in 3D.

1.12 Constraint Notation

The project uses generic constraint symbols first, then specific ones when needed.

Generic notation:

$$C = \{\text{type}, \text{value}\}$$

Typical scalar limits in 1D are:

- v_{max} : maximum admissible velocity;
- a_{max} : maximum admissible acceleration;
- j_{max} : maximum admissible jerk;
- s_{max} : maximum admissible snap.

When a constant-velocity sub-segment is required, the notation is:

$$x_{cv}^{start}, \quad x_{cv}^{end}, \quad v_{cv}$$

in 1D, and:

$$\mathbf{p}_{cv}^{start}, \quad \mathbf{p}_{cv}^{end}, \quad v_{cv}$$

in 3D.

The scalar v_{cv} denotes the requested constant speed magnitude. If direction matters in 3D, the corresponding constant velocity vector should be written as:

$$\mathbf{v}_{cv}$$

1.13 Reserved Orientation Notation

Orientation is not developed here, but the following symbols are reserved so that future documents remain coherent:

- Euler angles: ϕ, θ, ψ
- rotation matrix: $\mathbf{R}$
- quaternion: $\mathbf{q}$ or q , depending on the chosen convention

Because $\mathbf{q}$ may also be used for snap in some literature, future orientation documents must explicitly declare the chosen convention before use. Until then:

- use $\mathbf{p}^{(4)}(t)$ when referring to vector snap in potentially ambiguous contexts;
- use $\mathbf{R}, \phi, \theta, \psi$ for reserved orientation notation;
- avoid quaternion derivations for now.

1.14 Procedure For Later Documents

To keep the full library consistent, every later document should follow this practical sequence:

1. start from the notation defined here;
2. introduce only the symbols actually needed by that topic;
3. keep the 1D derivation as the base case;
4. extend to 3D by component-wise interpretation or vector form;
5. declare any new symbol locally if it is not already defined here.

This simple procedure is important for study: it avoids re-learning a different notation in every document.

1.15 Worked Symbolic Considerations

The notation can already be read in a symbolic way without numerical data.

Example 1, 1D endpoint problem:

$$x(t) : (x_0, v_0, a_0, j_0, s_0) \rightarrow (x_f, v_f, a_f, j_f, s_f)$$

Example 2, 3D endpoint problem:

$$\mathbf{p}(t) : (\mathbf{p}_0, \mathbf{v}_0, \mathbf{a}_0) \rightarrow (\mathbf{p}_f, \mathbf{v}_f, \mathbf{a}_f)$$

Example 3, normalized-time formulation:

$$x(\tau) = \sum_{i=0}^n \alpha_i \tau^i, \quad \tau \in [0, 1]$$

where α_i are the coefficients expressed in normalized time.

These examples are intentionally symbolic because the project prioritizes structural understanding before numerical substitution.

1.16 Open Questions

- **Resolved in BoundaryConditions:** for each polynomial degree, the minimal boundary-condition set is established there (necessary / possible / redundant classification, with a summary table for degrees 3, 5, 7, 9).
- **Resolved in TrajectoryConstraints:** whether 3D constraints should be imposed component-wise or on vector magnitude. This question also recurs in `NinthDegreePolynomialTheory`; it is intentionally not decided here, in isolation, but once the max/min velocity/acceleration/jerk/snap analysis (Req. 7–10) is developed in that document, with enough technical grounding to decide properly.
- **Open — to resolve during future orientation work** (see `04-future-topics.md`): should quaternion notation use q or $\mathbf{q}$ once snap notation is already present? No urgency until orientation is actually developed.

1.17 Related Documents

- `06-theory-library-roadmap.md`
- `NinthDegreePolynomialTheory`
- `BoundaryConditions`
- `PolynomialCoefficientDetermination`

2 Ninth-Degree Polynomial Theory

2.1 Intro And Scope

This document introduces the ninth-degree polynomial as the first complete reference case for the project. Its role is twofold:

- to define the polynomial in forms that are useful for derivation and implementation;
- to study the derivative chain from position up to the sixth derivative, so the later trajectory-planning documents can refer to a shared theoretical base.

The treatment starts in 1D, because that is the base case adopted by the project. The 3D case is then obtained by applying the same scalar law independently to the three Cartesian components.

This document focuses on Requirement 2 of `01-requirements.md`:

- canonical form and Horner form in the time domain;
- derivatives up to the sixth derivative;
- discussion of the usefulness of using more or fewer derivatives in trajectory planning.

2.2 Symbols Used

The notation follows `SymbolsAndNomenclature.md`. The main symbols used here are:

- t : absolute time;
- t_0 : initial time;
- t_f : final time;
- $T = t_f - t_0$: motion duration;
- $x(t)$: scalar position law in 1D;
- $a_0, a_1, \dots, a_9$: polynomial coefficients;
- S_0, S_f : initial and final state containers when endpoint data are grouped compactly;
- $v(t), a(t), j(t)$: velocity, acceleration, and jerk;
- $x^{(4)}(t), x^{(5)}(t), x^{(6)}(t)$: snap, crackle, and pop.

When normalized time is briefly mentioned, the symbol τ is used with $\tau \in [0, 1]$.

2.3 Why Degree Nine Is A Natural First Case

A ninth-degree polynomial has ten coefficients. This makes it especially useful as a first full study case because a trajectory segment can be determined by ten scalar conditions if the formulation requires them.

In the standard symmetric endpoint formulation, those ten conditions are naturally obtained by prescribing:

$$x_0, x_f, v_0, v_f, a_0, a_f, j_0, j_f, s_0, s_f$$

where s denotes snap.

This does not mean that every problem must always be posed this way. It means that degree nine is the first degree in this project that naturally accommodates endpoint information up to snap at both ends of a segment.

Using the state notation introduced in `SymbolsAndNomenclature.md`, the same endpoint specification may also be grouped as:

$$\mathcal{S}_0 = (x_0, v_0, a_0, j_0, s_0), \quad \mathcal{S}_f = (x_f, v_f, a_f, j_f, s_f)$$

This compact notation is useful when the full endpoint information is treated as a single conceptual object rather than as a list of separate scalar conditions.

2.4 Canonical Form In The Time Domain

The canonical form is the direct polynomial expansion in powers of time:

$$x(t) = a_0 + a_1t + a_2t^2 + a_3t^3 + a_4t^4 + a_5t^5 + a_6t^6 + a_7t^7 + a_8t^8 + a_9t^9$$

The same expression can be written compactly as:

$$x(t) = \sum_{i=0}^9 a_i t^i$$

This form is the most transparent one for theoretical work because every coefficient is visibly attached to its corresponding power of time.

2.4.1 Reading The Canonical Form

The canonical form is useful to read the structure of the motion law step by step:

1. the constant term a_0 is the value present even at $t = 0$;
2. the linear term a_1t contributes directly to initial slope;
3. higher powers progressively shape curvature and higher-order changes;
4. the highest-degree term a_9t^9 gives the polynomial its ninth-degree character.

This reading is qualitative, but it is educationally useful because it connects algebraic structure to motion behavior.

2.5 Horner Form In The Time Domain

The same polynomial can be rewritten in nested form:

$$x(t) = a_0 + t \left(a_1 + t \left(a_2 + t \left(a_3 + t \left(a_4 + t \left(a_5 + t \left(a_6 + t \left(a_7 + t \left(a_8 + t a_9 \right) \right) \right) \right) \right) \right) \right) \right)$$

Equivalent right-nested notation is:

$$x(t) = (((((((((a_9t + a_8)t + a_7)t + a_6)t + a_5)t + a_4)t + a_3)t + a_2)t + a_1)t + a_0)$$

These two expressions are algebraically identical. The difference is only in how the polynomial is organized for reading and evaluation.

2.5.1 How Horner Form Is Built

Starting from the canonical form:

$$x(t) = a_0 + a_1t + a_2t^2 + \cdots + a_9t^9$$

factor t progressively:

$$x(t) = a_0 + t(a_1 + a_2t + a_3t^2 + \cdots + a_9t^8)$$

then factor t again inside the parentheses:

$$x(t) = a_0 + t(a_1 + t(a_2 + a_3t + a_4t^2 + \cdots + a_9t^7))$$

and continue until the fully nested structure is obtained.

This is not a new polynomial. It is the same polynomial rewritten so that each step uses one multiplication by t and one addition.

2.6 Canonical Form Vs Horner Form

The two forms serve different purposes.

2.6.1 Canonical Form: Advantages

- It shows the polynomial degree immediately.
- It makes symbolic differentiation straightforward.
- It is convenient when setting up coefficient-determination equations.
- It is easier to inspect manually during theoretical study.
- It exposes clearly which coefficient multiplies which power of time.

2.6.2 Canonical Form: Limitations

- It is less efficient for repeated numerical evaluation.
- A direct implementation may compute many powers of t explicitly.
- Large powers of time can make numerical scaling less convenient.
- It is less natural for recursive evaluation.

2.6.3 Horner Form: Advantages

- It is computationally efficient for numerical evaluation.
- It avoids explicit computation of powers such as t^7 , t^8 , and t^9 .
- It reduces the number of multiplications needed.
- It is often better suited for implementation in software.
- It is easy to evaluate repeatedly for many time samples.

2.6.4 Horner Form: Limitations

- It hides the power-by-power structure of the polynomial.
- It is less intuitive when deriving equations by hand.
- It is less transparent for teaching coefficient meaning.
- It is not the most convenient form for symbolic derivations.

2.6.5 Practical Interpretation

For this project, the best attitude is not to treat the two forms as competitors, but as complementary tools:

- canonical form is the preferred form for theory, derivation, and boundary-condition reasoning;
- Horner form is the preferred form for efficient numerical evaluation in software.

This is an important educational point: different mathematical forms of the same object become useful at different stages of the project.

2.7 General Derivative Rule

If

$$x(t) = \sum_{i=0}^9 a_i t^i$$

then the k -th derivative is:

$$x^{(k)}(t) = \sum_{i=k}^9 \frac{i!}{(i-k)!} a_i t^{i-k}$$

This compact formula is useful because it shows the pattern behind all derivative expressions:

- every differentiation lowers the power of t by one;
- every differentiation multiplies the coefficient by the current exponent;
- terms of degree lower than k disappear in the k -th derivative.

2.8 Derivatives Of The Ninth-Degree Polynomial

Starting from

$$x(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5 + a_6 t^6 + a_7 t^7 + a_8 t^8 + a_9 t^9$$

the derivatives up to the sixth are the following.

2.8.1 First Derivative: Velocity

$$v(t) = \dot{x}(t) = a_1 + 2a_2 t + 3a_3 t^2 + 4a_4 t^3 + 5a_5 t^4 + 6a_6 t^5 + 7a_7 t^6 + 8a_8 t^7 + 9a_9 t^8$$

Velocity is the first quantity that describes how position changes in time. It is also the most immediate kinematic limit in many planning problems.

2.8.2 Second Derivative: Acceleration

$$a(t) = \ddot{x}(t) = 2a_2 + 6a_3 t + 12a_4 t^2 + 20a_5 t^3 + 30a_6 t^4 + 42a_7 t^5 + 56a_8 t^6 + 72a_9 t^7$$

Acceleration measures how velocity changes. It is central whenever smooth dynamic evolution is more important than position alone.

2.8.3 Third Derivative: Jerk

$$j(t) = x^{(3)}(t) = 6a_3 + 24a_4t + 60a_5t^2 + 120a_6t^3 + 210a_7t^4 + 336a_8t^5 + 504a_9t^6$$

Jerk is often introduced when acceleration continuity is not sufficient and sharper changes need to be controlled.

2.8.4 Fourth Derivative: Snap

$$x^{(4)}(t) = 24a_4 + 120a_5t + 360a_6t^2 + 840a_7t^3 + 1680a_8t^4 + 3024a_9t^5$$

Snap is especially relevant in a ninth-degree formulation because endpoint snap values can be included naturally among the ten scalar conditions.

2.8.5 Fifth Derivative: Crackle

$$x^{(5)}(t) = 120a_5 + 720a_6t + 2520a_7t^2 + 6720a_8t^3 + 15120a_9t^4$$

Crackle is less commonly used as a physical planning quantity, but it remains analytically useful because it is the derivative of snap.

2.8.6 Sixth Derivative: Pop

$$x^{(6)}(t) = 720a_6 + 5040a_7t + 20160a_8t^2 + 60480a_9t^3$$

Pop is the derivative of crackle. It is rarely a primary planning constraint, but it is part of the natural derivative chain of a ninth-degree polynomial and can help understand higher-order variation.

2.9 What Changes In 3D

The 3D case does not introduce a new scalar theory. It applies the same ninth-degree law to each Cartesian component:

$$\mathbf{p}(t) = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix}$$

with

$$x(t) = \sum_{i=0}^9 a_i^{(x)} t^i, \quad y(t) = \sum_{i=0}^9 a_i^{(y)} t^i, \quad z(t) = \sum_{i=0}^9 a_i^{(z)} t^i$$

The same derivative chain applies component-wise:

$$\dot{\mathbf{p}}(t), \quad \ddot{\mathbf{p}}(t), \quad \mathbf{p}^{(3)}(t), \quad \mathbf{p}^{(4)}(t), \quad \mathbf{p}^{(5)}(t), \quad \mathbf{p}^{(6)}(t)$$

So, from the viewpoint of this document:

- the 1D theory is the real mathematical core;
- the 3D extension is obtained by repeating that theory on three components;
- extra difficulty in 3D comes later from coupling choices, shared duration, and constraint interpretation, not from a different polynomial basis.

2.10 Why Fewer Derivatives May Be Enough

In many trajectory-planning problems, using fewer derivatives is entirely reasonable.

2.10.1 Position And Velocity Only

If the problem only requires reaching a target position with prescribed endpoint velocities, a lower-order description may be sufficient. This is useful when:

- the motion is simple;
- dynamic smoothness beyond velocity is not critical;
- the polynomial degree can be kept lower.

The advantage is simplicity. The limitation is that acceleration and higher-order behavior are left less controlled.

2.10.2 Up To Acceleration

Using position, velocity, and acceleration is common when smoother motion is needed. This usually gives a better compromise between control and complexity.

Acceleration-level reasoning is already very meaningful because:

- many mechanical effects are more closely related to acceleration than to position alone;
- acceleration continuity usually improves motion quality substantially;
- it avoids some abrupt changes that remain hidden if only velocity is considered.

2.10.3 Up To Jerk

Including jerk is often valuable when the transition smoothness itself becomes a design concern. Even without discussing actuators in detail, jerk remains educationally important because it captures how aggressively acceleration changes.

For this project, jerk is especially useful because it forms the bridge between basic trajectory geometry and genuinely smooth motion planning.

2.11 Why More Derivatives Can Be Useful

Studying derivatives beyond jerk is not always necessary, but it can still be useful for at least three reasons.

2.11.1 First Reason: Endpoint Specification Richness

Degree nine allows endpoint conditions up to snap at both ends. If snap is part of the formulation, then studying snap explicitly is not optional; it belongs to the definition of the segment itself.

2.11.2 Second Reason: Higher-Order Smoothness

Even if crackle and pop are not imposed as boundary conditions, they still describe how higher-order quantities evolve. This is useful when the goal is not only to obtain a valid trajectory, but to understand how smooth it really is.

2.11.3 Third Reason: Analytical Support For Later Constraints

Higher derivatives appear naturally when later documents analyze extrema:

- extrema of velocity are related to acceleration;
- extrema of acceleration are related to jerk;
- extrema of jerk are related to snap;
- extrema of snap are related to crackle.

So even if crackle and pop are not used as physical constraints, they remain mathematically useful.

2.12 Are Crackle And Pop Physically Necessary

Usually, no.

For most practical planning formulations, crackle and pop are not primary quantities to prescribe directly. In this project they are mainly useful because:

- they complete the derivative hierarchy of the ninth-degree polynomial;
- they help interpret the behavior of snap and higher-order smoothness;
- they support later analytical reasoning about maxima, minima, and continuity.

In other words:

- position, velocity, acceleration, and jerk are often directly meaningful;
- snap may become directly meaningful in high-smoothness formulations;
- crackle and pop are often more useful as analytical companions than as primary design targets.

2.13 The Special Role Of Snap In Degree Nine

Among the higher derivatives, snap has a special status in this project.

The reason is structural: a ninth-degree polynomial has ten coefficients, and the symmetric endpoint data set

$$\mathcal{S}_0 = (x_0, v_0, a_0, j_0, s_0), \quad \mathcal{S}_f = (x_f, v_f, a_f, j_f, s_f)$$

provides exactly ten scalar conditions.

This makes snap the highest derivative that fits naturally into the standard two-endpoint determination of a ninth-degree segment.

That is why this document studies derivatives beyond jerk, but still treats snap as the highest derivative with a particularly strong structural role.

2.14 A Practical Evaluation Procedure

When the polynomial must be evaluated numerically at many time instants, the following workflow is useful:

1. derive the equations in canonical form;
2. determine the coefficients from the chosen boundary conditions;
3. convert the polynomial to Horner form for repeated evaluation;
4. sample the motion over time to obtain position and derivative profiles.

This workflow matches the educational goal of the project:

- theory remains readable in canonical form;
- implementation remains efficient in Horner form.

2.15 Worked Symbolic Considerations

2.15.1 Example 1: Full Ninth-Degree Segment In 1D

Consider the symbolic segment

$$x(t) = \sum_{i=0}^9 a_i t^i$$

with endpoint data prescribed up to snap at both ends.

The key observation is not yet the solution for the coefficients, but the structural match:

- unknowns: $a_0, a_1, \dots, a_9$;
- scalar equations: initial and final values of position, velocity, acceleration, jerk, and snap.

This is why the ninth-degree polynomial is a natural theoretical anchor for the rest of the project.

2.15.2 Example 2: Why Higher Derivatives Matter For Extrema

Suppose a later document needs the maximum snap value. Then the stationary points of snap are found from:

$$\frac{d}{dt} \left(x^{(4)}(t) \right) = x^{(5)}(t) = 0$$

So crackle appears naturally even if it was never chosen as a planning constraint.

2.15.3 Example 3: 3D Interpretation

If

$$\mathbf{p}(t) = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix}$$

then a ninth-degree 3D segment is simply three synchronized ninth-degree scalar segments. The scalar theory does not change; only the bookkeeping grows.

2.16 Open Questions

- **Open — to resolve during the software design/implementation phase (src/):** should derivative profiles be evaluated directly from canonical formulas or from Horner-style recursive schemes? This is not a theory-library decision — both forms are proven equivalent above; the choice depends on measured performance in the actual C#/WPF implementation, not on the underlying math.
- **Resolved in TrajectoryConstraints:** whether limits should be interpreted component-wise or on vector magnitude when extending from 1D to 3D. This duplicates the same open question in `SymbolsAndNomenclature` — see the consolidated note there; it will be resolved once developed in `TrajectoryConstraints`, not decided here.
- **Open — to resolve during the application/UI design phase (Req. 15):** in the final application, which derivative levels should be visible by default in the charts, and which should remain optional?

2.17 Related Documents

- 06-theory-library-roadmap.md
- SymbolsAndNomenclature.md
- BoundaryConditions
- PolynomialCoefficientDetermination
- TrajectoryConstraints

3 Boundary Conditions

3.1 Intro And Scope

This document explains what boundary conditions are in the context of polynomial trajectory planning and lists the boundary conditions relevant to each polynomial degree currently in scope for the project: degree 3, 5, 7, and 9.

The goal of the document is educational before technical. It clarifies:

- what a boundary condition actually is;
- why the number of boundary conditions depends on the polynomial degree;
- how to distinguish necessary, possible, and redundant conditions;
- how the 1D reasoning extends to 3D.

This document covers the learning part of Requirement 3 in `01-requirements.md`.

3.2 Symbols Used

The notation follows `SymbolsAndNomenclature.md`. The main symbols used here are:

- t_0 : initial time;
- t_f : final time;
- $T = t_f - t_0$: motion duration;
- $x(t)$: scalar position law in 1D;
- $\mathcal{S}_0, \mathcal{S}_f$: initial and final state containers when endpoint data are grouped compactly;
- x_0, x_f : initial and final position;
- v_0, v_f : initial and final velocity;
- a_0, a_f : initial and final acceleration;
- j_0, j_f : initial and final jerk;
- s_0, s_f : initial and final snap.

When needed, the generic k -th derivative is written as $x^{(k)}(t)$.

3.3 What A Boundary Condition Is

A boundary condition is a prescribed value imposed on the trajectory or on one of its derivatives at a specific time, usually at the beginning or at the end of a segment.

Typical examples are:

$$x(t_0) = x_0, \quad x(t_f) = x_f$$

for position, or:

$$\dot{x}(t_0) = v_0, \quad \dot{x}(t_f) = v_f$$

for velocity.

So, in simple terms:

- a polynomial provides unknown coefficients;
- boundary conditions provide equations;
- when the number of independent equations matches the number of unknown coefficients, the polynomial can be determined uniquely.

This is the key idea behind the whole document.

3.4 Why The Number Of Boundary Conditions Depends On Degree

A polynomial of degree n has $n + 1$ coefficients:

$$x(t) = a_0 + a_1t + a_2t^2 + \cdots + a_nt^n$$

The unknowns are therefore:

$$a_0, a_1, a_2, \dots, a_n$$

which are $n + 1$ scalar unknowns.

If the duration T is already known and fixed, then the standard coefficient-determination problem requires:

$$n + 1$$

independent scalar boundary conditions.

This immediately gives the count for the degrees currently in scope:

- degree 3: 4 conditions;
- degree 5: 6 conditions;
- degree 7: 8 conditions;
- degree 9: 10 conditions.

3.5 The Natural Symmetric Endpoint Pattern

For the odd degrees used in this project, there is a natural symmetric pattern based on matching the same derivative orders at the start and at the end of the segment.

The pattern is:

- degree 3: position and velocity at both ends;
- degree 5: position, velocity, and acceleration at both ends;
- degree 7: position, velocity, acceleration, and jerk at both ends;
- degree 9: position, velocity, acceleration, jerk, and snap at both ends.

This pattern is natural because it gives exactly the required number of scalar conditions:

$$2 \times \frac{n + 1}{2} = n + 1$$

for odd degree n .

This is not the only possible formulation, but it is the most direct and educational one for this project.

3.6 Three Condition Classes

To keep the terminology consistent across the project, boundary conditions are divided into three classes.

3.6.1 Necessary Conditions

Necessary conditions are the conditions that must be specified in the chosen formulation to determine the polynomial uniquely.

In this document, the default formulation is:

- fixed polynomial degree;
- fixed start and end times;
- endpoint conditions only;
- symmetric use of derivative orders at the two endpoints.

Under this formulation, the necessary conditions are the ones listed degree by degree in the next sections.

3.6.2 Possible Conditions

Possible conditions are conditions that can be used in broader or alternative formulations, but are not required in the default formulation.

Examples:

- replacing one endpoint derivative condition with another independent condition;
- prescribing an interior-point condition;
- using higher-order derivative data while freeing another parameter such as duration;
- using a non-symmetric formulation between start and end conditions.

So “possible” does not mean “needed here”. It means “admissible in some extended formulation”.

3.6.3 Redundant Conditions

Redundant conditions are extra independent conditions beyond what the fixed-degree, fixed-duration polynomial can satisfy uniquely.

If a degree- n polynomial with fixed duration already has $n + 1$ independent scalar conditions, then adding another independent condition makes the problem overconstrained.

In that situation, one of the following happens:

- the system has no exact solution;
- one of the conditions is not actually independent;
- the formulation must be changed, for example by increasing degree or adding free parameters.

This is why the distinction between possible and redundant is important:

- possible means usable in some reformulated problem;
- redundant means extra within the current fixed formulation.

3.7 Condition Catalogue In 1D

Before listing the degrees one by one, it is useful to define the basic catalogue of endpoint conditions in 1D:

$$x_0, x_f, v_0, v_f, a_0, a_f, j_0, j_f, s_0, s_f$$

These correspond to:

- position at start and end;
- velocity at start and end;
- acceleration at start and end;
- jerk at start and end;
- snap at start and end.

For the degrees currently in scope, this catalogue is sufficient for the main endpoint-based theory.

When a full endpoint condition set is treated as a compact object, the state notation from `SymbolsAndNomenclature.md` may also be used. For example, in the degree-9 case:

$$\mathcal{S}_0 = (x_0, v_0, a_0, j_0, s_0), \quad \mathcal{S}_f = (x_f, v_f, a_f, j_f, s_f)$$

This notation is particularly useful when discussing complete endpoint data sets without overloading the snap symbol s .

3.8 Degree 3

3.8.1 Polynomial Structure

A cubic polynomial has four coefficients:

$$x(t) = a_0 + a_1t + a_2t^2 + a_3t^3$$

Therefore it needs four independent scalar conditions in the default formulation.

3.8.2 Necessary Conditions

The natural necessary set is:

$$x_0, x_f, v_0, v_f$$

This means:

$$x(t_0) = x_0, \quad x(t_f) = x_f, \quad \dot{x}(t_0) = v_0, \quad \dot{x}(t_f) = v_f$$

3.8.3 Possible Conditions

Possible conditions in broader formulations may include:

- acceleration values at one or both endpoints;
- jerk values at one or both endpoints;
- an interior position value;
- unknown duration combined with a different choice of endpoint conditions.

These conditions are possible only if the full formulation is changed so that the total number of independent unknowns and equations remains balanced.

3.8.4 Redundant Conditions

In the default cubic formulation with fixed duration, the following types of additions are redundant:

- prescribing a_0 or a_f in addition to x_0, x_f, v_0, v_f ;
- prescribing j_0 or j_f in addition to the same set;
- adding any extra independent endpoint condition beyond the original four.

So the basic educational message is simple: a cubic segment naturally controls position and velocity, not the full higher-order hierarchy.

3.9 Degree 5

3.9.1 Polynomial Structure

A quintic polynomial has six coefficients:

$$x(t) = a_0 + a_1t + a_2t^2 + a_3t^3 + a_4t^4 + a_5t^5$$

Therefore it needs six independent scalar conditions in the default formulation.

3.9.2 Necessary Conditions

The natural necessary set is:

$$x_0, x_f, v_0, v_f, a_0, a_f$$

This means:

$$x(t_0) = x_0, \quad x(t_f) = x_f$$

$$\dot{x}(t_0) = v_0, \quad \dot{x}(t_f) = v_f$$

$$\ddot{x}(t_0) = a_0, \quad \ddot{x}(t_f) = a_f$$

3.9.3 Possible Conditions

Possible conditions in extended formulations may include:

- jerk at one or both endpoints;
- one or more interior conditions;
- duration treated as an unknown;
- a non-symmetric distribution of endpoint derivative conditions.

3.9.4 Redundant Conditions

In the default quintic formulation with fixed duration, the following become redundant if added independently:

- j_0, j_f ;
- any snap endpoint condition;
- any extra independent condition beyond the six already listed.

The quintic therefore extends the cubic case by naturally controlling acceleration at the boundaries.

3.10 Degree 7

3.10.1 Polynomial Structure

A seventh-degree polynomial has eight coefficients:

$$x(t) = a_0 + a_1t + a_2t^2 + a_3t^3 + a_4t^4 + a_5t^5 + a_6t^6 + a_7t^7$$

Therefore it needs eight independent scalar conditions in the default formulation.

3.10.2 Necessary Conditions

The natural necessary set is:

$$x_0, x_f, v_0, v_f, a_0, a_f, j_0, j_f$$

This means that position, velocity, acceleration, and jerk are prescribed at both endpoints.

3.10.3 Possible Conditions

Possible conditions in broader formulations may include:

- snap at one or both endpoints;
- interior constraints;
- alternative mixes of endpoint data;
- duration or other parameters promoted to unknowns.

3.10.4 Redundant Conditions

In the default seventh-degree formulation with fixed duration, the following become redundant if added independently:

- s_0, s_f ;
- any extra endpoint condition beyond the eight already used;
- any other independent scalar condition that does not replace an existing one.

The educational interpretation is that the seventh-degree polynomial is the first degree in this project that naturally includes jerk at both ends.

3.11 Degree 9

3.11.1 Polynomial Structure

A ninth-degree polynomial has ten coefficients:

$$x(t) = a_0 + a_1t + a_2t^2 + a_3t^3 + a_4t^4 + a_5t^5 + a_6t^6 + a_7t^7 + a_8t^8 + a_9t^9$$

Therefore it needs ten independent scalar conditions in the default formulation.

3.11.2 Necessary Conditions

The natural necessary set is:

$$x_0, x_f, v_0, v_f, a_0, a_f, \dot{j}_0, \dot{j}_f, s_0, s_f$$

This means that position, velocity, acceleration, jerk, and snap are prescribed at both endpoints.

3.11.3 Possible Conditions

Possible conditions in broader formulations may include:

- crackle-related information if the problem is reformulated;
- interior-point constraints;
- asymmetric endpoint specifications;
- duration included among the unknowns;
- optimization-based or approximate fitting formulations.

3.11.4 Redundant Conditions

In the default ninth-degree formulation with fixed duration, the following become redundant if added independently:

- any additional crackle or pop endpoint condition;
- any extra interior scalar condition that is imposed without releasing another condition or adding unknowns;
- any extra independent scalar condition beyond the ten already used.

This is why snap has a special role in the degree-nine case: it is the highest derivative that still fits naturally into the standard symmetric endpoint formulation.

3.12 Summary Table For 1D

The degree-by-degree classification can be summarized as follows.

Degree	Coefficients	Necessary conditions in the default formulation	Examples of possible extra conditions in broader formulations	Redundant in the default fixed formulation
3	4	x_0, x_f, v_0, v_f	a_0, a_f , interior position, unknown duration	any fifth independent condition
5	6	$x_0, x_f, v_0, v_f, a_0, a_f$	j_0, j_f , interior conditions, unknown duration	any seventh independent condition
7	8	$x_0, x_f, v_0, v_f, a_0, a_f, j_0, j_f$	s_0, s_f , interior conditions, unknown duration	any ninth independent condition
9	10	$x_0, x_f, v_0, v_f, a_0, a_f, j_0, j_f, s_0, s_f$	crackle-related data, interior conditions, unknown duration	any eleventh independent condition

This table should be read carefully:

- the “necessary” column refers only to the standard formulation adopted as default in this project;
- the “possible” column does not mean those conditions are added on top unchanged;
- the “redundant” column means “redundant unless the formulation itself is changed”.

3.13 Independence Matters As Much As Count

Counting conditions is necessary, but not sufficient. The conditions must also be independent.

For example, a degree-5 polynomial needs six scalar conditions, but not every group of six conditions is equally well chosen. If two conditions are linearly dependent in the coefficient system, they do not provide six independent equations.

So the complete rule is:

- correct number of conditions;
- independence of those conditions;
- consistency with the chosen formulation.

This point will become especially important in `PolynomialCoefficientDetermination`.

3.14 What Changes In 3D

The logic does not change in 3D. What changes is that the scalar 1D problem is applied to three Cartesian components.

If the three coordinates are treated independently, then each axis has its own set of scalar boundary conditions:

$$x_0, x_f, v_{x,0}, v_{x,f}, \dots$$

$$y_0, y_f, v_{y,0}, v_{y,f}, \dots$$

$$z_0, z_f, v_{z,0}, v_{z,f}, \dots$$

In compact vector notation, the corresponding endpoint conditions are:

$$\mathbf{p}_0, \mathbf{p}_f, \mathbf{v}_0, \mathbf{v}_f, \mathbf{a}_0, \mathbf{a}_f, \mathbf{j}_0, \mathbf{j}_f, \mathbf{p}_0^{(4)}, \mathbf{p}_f^{(4)}$$

The educational idea is still the same:

- first understand the scalar case;
- then apply the same rule component-wise in 3D.

The main difference in 3D is therefore not the nature of the boundary conditions, but the bookkeeping and the later interpretation of multi-axis coordination.

3.15 Relation To The Data Model

02-data-model.md now defines `BoundaryConditions` as:

$$\text{BoundaryConditions} = \{\text{necessaryConditions}[], \text{possibleConditions}[], \text{redundantConditions}[]\}$$

This is the degree-parametrized, classified structure that this document’s necessary/possible/redundant classification calls for — the revision has been carried out, replacing the earlier fixed set of ten scalar fields (`startPosition/endPosition`, etc.) that matched only the degree-9 default formulation.

The three arrays hold:

- `necessaryConditions[]`: the conditions required by the chosen degree and formulation;
- `possibleConditions[]`: conditions usable in alternative formulations;
- `redundantConditions[]`: conditions that exceed the default fixed formulation.

What remains open is not *whether* to revise the structure (done), but the internal representation of each individual condition within those arrays — see the corresponding open question below, to be resolved together with `PolynomialCoefficientDetermination`.

In later implementation, each condition will likely need a more explicit internal representation, for example:

- derivative order;
- endpoint identifier;
- symbolic or numeric value.

3.16 Worked Symbolic Considerations

3.16.1 Example 1: Cubic Segment

For a cubic segment with fixed duration, the unknown coefficients are:

$$a_0, a_1, a_2, a_3$$

The natural endpoint conditions are:

$$x_0, x_f, v_0, v_f$$

This gives four independent scalar equations for four unknowns.

If one now adds a_0 and a_f as prescribed endpoint accelerations without changing the formulation, the cubic problem becomes overconstrained.

3.16.2 Example 2: Quintic Segment

For a quintic segment, the unknown coefficients are:

$$a_0, a_1, a_2, a_3, a_4, a_5$$

The natural endpoint conditions are:

$$x_0, x_f, v_0, v_f, a_0, a_f$$

This is why quintic polynomials are often associated with endpoint control up to acceleration.

3.16.3 Example 3: Ninth-Degree Segment

For a ninth-degree segment, the natural symmetric endpoint set is:

$$\mathcal{S}_0 = (x_0, v_0, a_0, j_0, s_0), \quad \mathcal{S}_f = (x_f, v_f, a_f, j_f, s_f)$$

This gives ten scalar conditions, matching the ten coefficients of the polynomial exactly.

That exact structural match is one of the main reasons why the degree-nine case is the central reference of the project.

3.17 Open Questions

- **Resolved:** whether `02-data-model.md`'s `BoundaryConditions` structure should be revised to the degree-parametrized, classified form — it has been, see *Relation To The Data Model*. **Still open, deferred to PolynomialCoefficientDetermination:** the internal representation of each individual condition within `necessaryConditions[]` / `possibleConditions[]` / `redundantConditions[]` (by derivative order vs. a more descriptive enum-based type) — this is `PolynomialCoefficientDetermination`'s own Open Question on the same topic, not yet decided there either.
- **Open — to resolve if/when alternative formulations are introduced** (interior points, optimization-based fitting): should the project keep the same necessary/possible/redundant naming, or add a fourth category for “active in the current variant”? No urgency yet, since the project still uses the fixed default formulation.
- **Open — to resolve during the application/UI design phase** (Req. 15): for 3D trajectories, should the user interface expose boundary conditions axis by axis, or primarily in vector form?

3.18 Related Documents

- `06-theory-library-roadmap.md`
- `SymbolsAndNomenclature.md`
- `NinthDegreePolynomialTheory.md`
- `PolynomialCoefficientDetermination`
- `02-data-model.md`

4 Polynomial Coefficient Determination

4.1 Intro And Scope

This document explains how to determine the coefficients of a polynomial trajectory once the necessary boundary data has been specified.

It covers two connected themes:

- the technical procedure required by Requirement 3 in `01-requirements.md`;
- the learning-oriented comparison between absolute time and normalized time required by Requirement 4.

The discussion starts in 1D, because that is the conceptual base of the project. The 3D case is then obtained by applying the same scalar procedure to the Cartesian components.

The focus is on degrees 3, 5, 7, and 9, which are the degrees currently in scope for the project.

4.2 Symbols Used

The notation follows `SymbolsAndNomenclature.md`. The main symbols used here are:

- t : absolute time;
- t_0 : initial time;
- t_f : final time;
- $T = t_f - t_0$: segment duration;
- $\tau = \frac{t-t_0}{T}$: normalized time;
- $x(t)$: scalar position law in 1D written in absolute time;
- $x(\tau)$: scalar position law in normalized time;
- n : polynomial degree;
- a_i : coefficients of the polynomial when written in absolute time;
- α_i : coefficients of the polynomial when written in normalized time;
- $\mathcal{S}_0, \mathcal{S}_f$: endpoint state containers when compact notation is useful.

When derivatives are written in the normalized variable, the notation

$$\frac{d^k x}{d\tau^k}$$

is preferred in order to keep it clearly distinct from derivatives with respect to absolute time.

4.3 Problem Statement

The coefficient-determination problem can be stated as follows.

Given:

- a polynomial degree n ;
- a fixed segment duration T ;
- the necessary boundary conditions for that degree;

determine the polynomial coefficients so that all required endpoint conditions are satisfied exactly.

For the degrees currently used in the project, the default endpoint pattern is:

- degree 3: position and velocity at both ends;
- degree 5: position, velocity, and acceleration at both ends;
- degree 7: position, velocity, acceleration, and jerk at both ends;
- degree 9: position, velocity, acceleration, jerk, and snap at both ends.

This means that coefficient determination is always a square linear problem in the default formulation:

- degree 3: 4 unknown coefficients, 4 scalar equations;
- degree 5: 6 unknown coefficients, 6 scalar equations;
- degree 7: 8 unknown coefficients, 8 scalar equations;
- degree 9: 10 unknown coefficients, 10 scalar equations.

4.4 Core Idea

A polynomial trajectory has unknown coefficients. Each boundary condition produces one scalar equation in those coefficients.

So the practical workflow is always:

1. choose the polynomial degree;
2. write the polynomial and its derivatives;
3. evaluate the required derivatives at the endpoints;
4. assemble the linear system;
5. solve for the coefficients;
6. verify the result by substitution.

The mathematical heart of the method is therefore not a special trick, but a structured linear system.

4.5 Absolute-Time Formulation

4.5.1 General Polynomial

In absolute time, the polynomial is written as:

$$x(t) = \sum_{i=0}^n a_i t^i$$

with unknown coefficient vector:

$$\mathbf{a} = [a_0 \quad a_1 \quad \dots \quad a_n]^T$$

4.5.2 General Derivative Rule

The k -th derivative is:

$$x^{(k)}(t) = \sum_{i=k}^n \frac{i!}{(i-k)!} a_i t^{i-k}$$

This formula is enough to generate every row of the coefficient-determination system.

4.5.3 Derivative Row Operators

For compact notation, define the row vector $\mathbf{D}_k(t)$ as the coefficient row that maps $\mathbf{a}$ to the k -th derivative value at time t :

$$x^{(k)}(t) = \mathbf{D}_k(t) \mathbf{a}$$

For degree 9, the first five rows are:

$$\mathbf{D}_0(t) = [1 \quad t \quad t^2 \quad t^3 \quad t^4 \quad t^5 \quad t^6 \quad t^7 \quad t^8 \quad t^9]$$

$$\mathbf{D}_1(t) = [0 \quad 1 \quad 2t \quad 3t^2 \quad 4t^3 \quad 5t^4 \quad 6t^5 \quad 7t^6 \quad 8t^7 \quad 9t^8]$$

$$\mathbf{D}_2(t) = [0 \quad 0 \quad 2 \quad 6t \quad 12t^2 \quad 20t^3 \quad 30t^4 \quad 42t^5 \quad 56t^6 \quad 72t^7]$$

$$\mathbf{D}_3(t) = [0 \quad 0 \quad 0 \quad 6 \quad 24t \quad 60t^2 \quad 120t^3 \quad 210t^4 \quad 336t^5 \quad 504t^6]$$

$$\mathbf{D}_4(t) = [0 \quad 0 \quad 0 \quad 0 \quad 24 \quad 120t \quad 360t^2 \quad 840t^3 \quad 1680t^4 \quad 3024t^5]$$

These rows are directly derived from `NinthDegreePolynomialTheory.md`.

4.6 Degree-Specific Linear Systems In Absolute Time

The default square systems for the degrees in scope are obtained by stacking the derivative rows at t_0 and t_f .

4.6.1 Degree 3

For

$$x(t) = a_0 + a_1t + a_2t^2 + a_3t^3$$

use:

$$\mathbf{a}_3 = [a_0 \quad a_1 \quad a_2 \quad a_3]^T$$

and:

$$\mathbf{b}_3 = [x_0 \quad v_0 \quad x_f \quad v_f]^T$$

The system is:

$$\mathbf{A}_3^{abs} \mathbf{a}_3 = \mathbf{b}_3$$

with:

$$\mathbf{A}_3^{abs} = \begin{bmatrix} \mathbf{D}_0(t_0) \\ \mathbf{D}_1(t_0) \\ \mathbf{D}_0(t_f) \\ \mathbf{D}_1(t_f) \end{bmatrix}$$

where each row is truncated to degree 3.

4.6.2 Degree 5

Use:

$$\mathbf{b}_5 = [x_0 \quad v_0 \quad a_0 \quad x_f \quad v_f \quad a_f]^T$$

and:

$$\mathbf{A}_5^{abs} = \begin{bmatrix} \mathbf{D}_0(t_0) \\ \mathbf{D}_1(t_0) \\ \mathbf{D}_2(t_0) \\ \mathbf{D}_0(t_f) \\ \mathbf{D}_1(t_f) \\ \mathbf{D}_2(t_f) \end{bmatrix}$$

again truncated to degree 5.

4.6.3 Degree 7

Use:

$$\mathbf{b}_7 = [x_0 \quad v_0 \quad a_0 \quad j_0 \quad x_f \quad v_f \quad a_f \quad j_f]^T$$

and:

$$\mathbf{A}_7^{abs} = \begin{bmatrix} \mathbf{D}_0(t_0) \\ \mathbf{D}_1(t_0) \\ \mathbf{D}_2(t_0) \\ \mathbf{D}_3(t_0) \\ \mathbf{D}_0(t_f) \\ \mathbf{D}_1(t_f) \\ \mathbf{D}_2(t_f) \\ \mathbf{D}_3(t_f) \end{bmatrix}$$

truncated to degree 7.

4.6.4 Degree 9

Use the compact state notation:

$$\mathcal{S}_0 = (x_0, v_0, a_0, j_0, s_0), \quad \mathcal{S}_f = (x_f, v_f, a_f, j_f, s_f)$$

or, equivalently, the endpoint vector:

$$\mathbf{b}_9 = [x_0 \quad v_0 \quad a_0 \quad j_0 \quad s_0 \quad x_f \quad v_f \quad a_f \quad j_f \quad s_f]^T$$

Then:

$$\mathbf{A}_9^{abs} \mathbf{a}_9 = \mathbf{b}_9$$

with:

$$\mathbf{A}_9^{abs} = \begin{bmatrix} \mathbf{D}_0(t_0) \\ \mathbf{D}_1(t_0) \\ \mathbf{D}_2(t_0) \\ \mathbf{D}_3(t_0) \\ \mathbf{D}_4(t_0) \\ \mathbf{D}_0(t_f) \\ \mathbf{D}_1(t_f) \\ \mathbf{D}_2(t_f) \\ \mathbf{D}_3(t_f) \\ \mathbf{D}_4(t_f) \end{bmatrix}$$

This is the direct absolute-time coefficient-determination system for the degree-9 case.

4.7 Step-By-Step Procedure In Absolute Time

The practical procedure in absolute time is the same for every supported degree.

1. Choose the degree n and identify the necessary endpoint conditions from `BoundaryConditions.md`.
2. Write the polynomial $x(t)$ in canonical form with unknown coefficients.
3. Compute the derivatives needed by that degree.
4. Evaluate those derivatives at t_0 and t_f .
5. Assemble the matrix $\mathbf{A}^{abs}$ and the right-hand side vector $\mathbf{b}$.
6. Solve the linear system for the coefficient vector $\mathbf{a}$.
7. Substitute the solution back into the polynomial and verify all endpoint conditions.

Conceptually, this is simple. In practice, the algebra becomes longer as the degree increases.

4.8 Why Normalized Time Is Worth Introducing

Absolute time works directly, but it is not always the most convenient variable.

When the segment duration is fixed, it is often useful to define:

$$\tau = \frac{t - t_0}{T}, \quad \tau \in [0, 1]$$

and write the same segment as:

$$x(\tau) = \sum_{i=0}^n \alpha_i \tau^i$$

The important idea is that:

- t belongs to the physical time axis;
- τ belongs to a normalized local segment axis;
- the two descriptions represent the same trajectory segment, but with different coefficients.

If $t_0 \neq 0$, the sets $\{a_i\}$ and $\{\alpha_i\}$ are not related by a simple one-to-one scaling. The normalized representation is centered on the local interval $[0, 1]$, whereas the absolute-time representation is written directly in powers of the global variable t .

4.9 Derivative Scaling Between t And τ

Because

$$\tau = \frac{t - t_0}{T}$$

we have:

$$\frac{d\tau}{dt} = \frac{1}{T}$$

Therefore:

$$\frac{dx}{dt} = \frac{dx}{d\tau} \frac{d\tau}{dt} = \frac{1}{T} \frac{dx}{d\tau}$$

and, more generally:

$$\frac{d^k x}{dt^k} = \frac{1}{T^k} \frac{d^k x}{d\tau^k}$$

Equivalently:

$$\frac{d^k x}{d\tau^k} = T^k \frac{d^k x}{dt^k}$$

This scaling rule is the key bridge between physical endpoint data and the normalized-time coefficient system.

4.10 Degree-9 System In Normalized Time

For degree 9, write:

$$x(\tau) = \alpha_0 + \alpha_1 \tau + \alpha_2 \tau^2 + \alpha_3 \tau^3 + \alpha_4 \tau^4 + \alpha_5 \tau^5 + \alpha_6 \tau^6 + \alpha_7 \tau^7 + \alpha_8 \tau^8 + \alpha_9 \tau^9$$

with coefficient vector:

$$\boldsymbol{\alpha} = [\alpha_0 \quad \alpha_1 \quad \alpha_2 \quad \alpha_3 \quad \alpha_4 \quad \alpha_5 \quad \alpha_6 \quad \alpha_7 \quad \alpha_8 \quad \alpha_9]^T$$

At $\tau = 0$, the first five endpoint equations become:

$$x(0) = \alpha_0 = x_0$$

$$\frac{dx}{d\tau}(0) = \alpha_1 = T v_0$$

$$\frac{d^2 x}{d\tau^2}(0) = 2\alpha_2 = T^2 a_0$$

$$\frac{d^3 x}{d\tau^3}(0) = 6\alpha_3 = T^3 j_0$$

$$\frac{d^4x}{d\tau^4}(0) = 24\alpha_4 = T^4s_0$$

At $\tau = 1$, the remaining equations are:

$$x(1) = \sum_{i=0}^9 \alpha_i = x_f$$

$$\frac{dx}{d\tau}(1) = \alpha_1 + 2\alpha_2 + 3\alpha_3 + 4\alpha_4 + 5\alpha_5 + 6\alpha_6 + 7\alpha_7 + 8\alpha_8 + 9\alpha_9 = Tv_f$$

$$\frac{d^2x}{d\tau^2}(1) = 2\alpha_2 + 6\alpha_3 + 12\alpha_4 + 20\alpha_5 + 30\alpha_6 + 42\alpha_7 + 56\alpha_8 + 72\alpha_9 = T^2a_f$$

$$\frac{d^3x}{d\tau^3}(1) = 6\alpha_3 + 24\alpha_4 + 60\alpha_5 + 120\alpha_6 + 210\alpha_7 + 336\alpha_8 + 504\alpha_9 = T^3j_f$$

$$\frac{d^4x}{d\tau^4}(1) = 24\alpha_4 + 120\alpha_5 + 360\alpha_6 + 840\alpha_7 + 1680\alpha_8 + 3024\alpha_9 = T^4s_f$$

So the normalized system is:

$$\mathbf{A}_9^{norm} \boldsymbol{\alpha} = \mathbf{b}_9^{norm}$$

with:

$$\mathbf{A}_9^{norm} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 6 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 24 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 0 & 0 & 2 & 6 & 12 & 20 & 30 & 42 & 56 & 72 \\ 0 & 0 & 0 & 6 & 24 & 60 & 120 & 210 & 336 & 504 \\ 0 & 0 & 0 & 0 & 24 & 120 & 360 & 840 & 1680 & 3024 \end{bmatrix}$$

and:

$$\mathbf{b}_9^{norm} = [x_0 \quad Tv_0 \quad T^2a_0 \quad T^3j_0 \quad T^4s_0 \quad x_f \quad Tv_f \quad T^2a_f \quad T^3j_f \quad T^4s_f]^T$$

This is one of the main practical advantages of normalized time: for a fixed degree, the matrix is constant and can be reused for every segment.

4.11 How The Lower Degrees Fit The Same Normalized Pattern

The same normalized-time logic applies to every supported degree.

4.11.1 Degree 3

Use:

$$\mathbf{b}_3^{norm} = [x_0 \quad Tv_0 \quad x_f \quad Tv_f]^T$$

with a constant 4×4 normalized matrix.

4.11.2 Degree 5

Use:

$$\mathbf{b}_5^{norm} = [x_0 \quad Tv_0 \quad T^2a_0 \quad x_f \quad Tv_f \quad T^2a_f]^T$$

with a constant 6×6 normalized matrix.

4.11.3 Degree 7

Use:

$$\mathbf{b}_7^{norm} = [x_0 \quad Tv_0 \quad T^2a_0 \quad T^3j_0 \quad x_f \quad Tv_f \quad T^2a_f \quad T^3j_f]^T$$

with a constant 8×8 normalized matrix.

4.11.4 Degree 9

Use the degree-9 system written in the previous section.

So the pattern is stable:

- the normalized matrix depends only on degree;
- the right-hand side contains the physical endpoint data scaled by powers of T .

4.12 Absolute Time Vs Normalized Time

The two formulations represent the same trajectory, but they are useful in different ways.

4.12.1 Absolute Time: Main Advantages

- It keeps the polynomial directly expressed in the physical variable t .
- It makes the final law immediately readable in the original time axis.
- It avoids introducing an extra variable.
- It is conceptually straightforward when $t_0 = 0$ and durations are simple.

4.12.2 Absolute Time: Main Limitations

- The system matrix changes with t_0 and t_f .
- Large or awkward time values generate large powers such as t_f^8 or t_f^9 .
- The algebra becomes less uniform from one segment to another.
- Numerical conditioning can become worse when absolute times are large.

4.12.3 Normalized Time: Main Advantages

- The segment always lives on the same interval $[0, 1]$.
- For a given degree, the system matrix is constant and reusable.
- The formulation is usually cleaner for symbolic derivation.
- Numerical scaling is often better because the variable does not grow beyond the local interval.
- Multi-segment implementations become more uniform because every segment uses the same local coordinate.

4.12.4 Normalized Time: Main Limitations

- Physical derivatives must be rescaled by powers of T .
- The coefficients α_i are not the same as the absolute-time coefficients a_i .
- If a final expression explicitly in powers of t is required, an extra conversion step is needed.
- The method introduces one extra conceptual layer for beginners.

4.12.5 Practical Interpretation For This Project

For this project, the most balanced approach is:

- use normalized time as the main working variable for coefficient determination;
- keep absolute time as the interpretation layer in which physical durations and derivative values are specified;
- evaluate the final trajectory numerically either in normalized time or, if needed, after conversion to absolute time.

This keeps the derivation clean without losing contact with physical meaning.

4.13 Recommended Determination Strategy

The following strategy is recommended for the project.

1. Express the trajectory segment in normalized time.
2. Build the constant normalized matrix associated with the chosen degree.
3. Assemble the right-hand side from the physical endpoint data scaled by T^k .
4. Solve for the normalized coefficients α_i .
5. Evaluate the trajectory and its derivatives using τ as the local segment variable.
6. Convert to an absolute-time polynomial only if that representation is explicitly needed.

This recommendation is especially useful for software implementation because it separates:

- physical inputs;
- degree-dependent matrix structure;
- reusable numerical evaluation.

4.14 Independence And Solvability

Counting equations is not enough. The boundary conditions must also be independent, and the matrix must be invertible.

So a well-posed coefficient-determination problem requires:

- the correct number of scalar conditions for the chosen degree;
- independence of those conditions;
- a consistent time definition with $T \neq 0$.

If these conditions fail, the system may become:

- underdetermined, if too few independent conditions are supplied;
- overdetermined, if too many independent conditions are imposed;

- singular, if the conditions are formally counted correctly but not independent.

This connects directly to the necessary / possible / redundant classification introduced in `BoundaryConditions.md`.

4.15 Relation To The Data Model

`02-data-model.md` currently defines:

$$\text{Polynomial} = \{\text{degree}, \text{coefficients}, \text{boundaryConditions}\}$$

and:

$$\text{BoundaryConditions} = \{\text{necessaryConditions}[], \text{possibleConditions}[], \text{redundantConditions}[]\}$$

From the viewpoint of coefficient determination, this means:

- **degree** selects the size and structure of the determination system;
- **necessaryConditions[]** provides the data that actually feeds the square linear system;
- **possibleConditions[]** may become relevant only in alternative formulations;
- **redundantConditions[]** should not be injected into the default fixed-degree system unchanged.

So the coefficient-determination procedure is the first place where the theoretical classification of boundary conditions becomes operational.

A future implementation-oriented refinement will likely need each necessary condition to carry at least:

- derivative order;
- endpoint identifier;
- scalar or vector value;
- dimensional context (1D or 3D).

4.16 What Changes In 3D

The determination logic does not change in 3D. The same scalar problem is solved component-wise.

For example, in a degree-9 segment:

$$\mathbf{p}(t) = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix}$$

the project can determine:

- one coefficient vector for the x component;
- one coefficient vector for the y component;
- one coefficient vector for the z component.

If the duration T is shared by all three axes, then the same normalized matrix is reused three times with three different right-hand side vectors.

This is one of the strongest reasons for keeping the 1D theory as the base case: the 3D extension is structurally repetitive rather than conceptually new.

4.17 Worked Symbolic Considerations

4.17.1 Example 1: Degree 3

For a cubic segment with fixed duration, the unknown coefficients are:

$$a_0, a_1, a_2, a_3$$

The natural endpoint data are:

$$x_0, v_0, x_f, v_f$$

These four scalar conditions generate a 4×4 square system. If the system matrix is invertible, the four coefficients are determined uniquely.

4.17.2 Example 2: Degree 9 In Normalized Time

For the degree-9 case, the endpoint data may be grouped as:

$$\mathcal{S}_0 = (x_0, v_0, a_0, j_0, s_0), \quad \mathcal{S}_f = (x_f, v_f, a_f, j_f, s_f)$$

The normalized system then uses:

$$x_0, T v_0, T^2 a_0, T^3 j_0, T^4 s_0, x_f, T v_f, T^2 a_f, T^3 j_f, T^4 s_f$$

as right-hand side data. The matrix stays constant; only the scaled endpoint vector changes from segment to segment.

4.17.3 Example 3: Why Normalized Time Helps In Multi-Segment Work

Suppose two trajectory segments have the same polynomial degree but different durations. In absolute time, their matrices are different because their endpoint times are different.

In normalized time, both segments use the same coefficient matrix. Only the right-hand side changes through the powers of each segment duration:

- one segment uses T_1, T_1^2, T_1^3 , and so on;
- the other uses T_2, T_2^2, T_2^3 , and so on.

This is mathematically cleaner and implementation-friendly.

4.18 Open Questions

- **Open — to resolve during software implementation (src/):** should the application store only normalized coefficients internally, or also keep an absolute-time coefficient expansion for inspection/export?
- **Deferred to the later constraints/blending documents:** when a formulation moves beyond the default necessary-condition set, should the software solve a restructured exact system, or transition to constrained/optimization-based formulations?
- **Open — to resolve during API/data-structure design:** should each condition within `necessaryConditions[]` / `possibleConditions[]` / `redundantConditions[]` be stored as a generic condition record, or as degree-specific strongly typed structures for degrees 3, 5, 7, and 9? (The classified array structure itself is already in place in `02-data-model.md`; only the internal representation of each entry remains open. Same open question as in `BoundaryConditions.md` — not yet decided in either document.)

4.19 Related Documents

- 06-theory-library-roadmap.md
- SymbolsAndNomenclature.md
- NinthDegreePolynomialTheory.md
- BoundaryConditions.md
- TrajectoryConstraints
- 02-data-model.md

5 Trajectory Constraints

5.1 Intro And Scope

This document studies trajectory constraints for the polynomial trajectories currently in scope, with the ninth-degree polynomial as the main technical reference case.

It covers the requirement cluster formed by items 7, 8, 9, and 10 in 01-requirements.md:

- determination of maxima and minima for position, velocity, acceleration, jerk, and snap;
- procedure to impose a velocity limit;
- procedure to impose an acceleration limit;
- procedure to impose a jerk limit.

The treatment starts in 1D and then extends to 3D. In keeping with 03-assumptions.md, the discussion remains theoretical, symbolic-first, and independent of actuator dynamics or robot kinematics.

5.2 Symbols Used

The notation follows SymbolsAndNomenclature.md. The main symbols used here are:

- t : absolute time;
- t_0 : initial time;
- t_f : final time;
- $T = t_f - t_0$: segment duration;
- $\tau = \frac{t-t_0}{T}$: normalized time;
- $x(t)$: scalar position law in 1D;
- $v(t) = \dot{x}(t)$: velocity;
- $a(t) = \ddot{x}(t)$: acceleration;
- $j(t) = x^{(3)}(t)$: jerk;
- $s(t) = x^{(4)}(t)$: snap;
- $x^{(5)}(t)$: crackle;
- $\mathbf{p}(t)$, $\mathbf{v}(t)$, $\mathbf{a}(t)$, $\mathbf{j}(t)$, $\mathbf{p}^{(4)}(t)$: 3D position and derivative vectors;
- v_{max} : admissible velocity limit;
- a_{max} : admissible acceleration limit;
- j_{max} : admissible jerk limit.

Unless stated otherwise, all extrema are searched over the closed interval:

$$t \in [t_0, t_f]$$

or, equivalently:

$$\tau \in [0, 1]$$

5.3 Constraint Context

In this project, a constraint is a condition imposed on the trajectory after the polynomial structure has been defined.

The most important kinematic constraints considered here are:

- bounds on velocity;
- bounds on acceleration;
- bounds on jerk.

Snap is included in the extrema analysis because Requirement 7 asks for it explicitly, even though there is no separate limit-imposition requirement for snap at this stage.

Position is also included in the extrema analysis, not because it is normally a “limit” in the same sense as velocity or acceleration, but because understanding where position reaches local or global extrema is part of understanding the trajectory profile.

5.4 Core Principle: Candidate Points For Extrema

For any scalar function $f(t)$ on a closed interval, the global maximum and minimum can only occur at:

- the interval endpoints;
- interior stationary points where $\frac{df}{dt} = 0$;
- interior points where the function is not differentiable.

Polynomial trajectories and their derivatives are smooth, so the third item does not apply here.

Therefore, the general constraint-analysis rule in this project is:

1. identify the scalar quantity of interest;
2. differentiate it;
3. solve the stationarity equation inside the interval;
4. evaluate the original quantity at all candidate points;
5. compare those values to obtain maxima and minima.

This rule is simple, but it drives almost the whole document.

5.5 Why Higher Derivatives Matter For Extrema

The derivative chain studied in `NinthDegreePolynomialTheory.md` now becomes operational.

For a degree-9 polynomial:

- extrema of position depend on velocity;
- extrema of velocity depend on acceleration;
- extrema of acceleration depend on jerk;
- extrema of jerk depend on snap;
- extrema of snap depend on crackle.

So the higher derivatives are not only abstract quantities. They are exactly the tools needed to locate the candidate extrema of the lower-order profiles.

5.6 Position Extrema In 1D

5.6.1 General Rule

To find maxima and minima of position $x(t)$ over $[t_0, t_f]$, solve:

$$v(t) = \dot{x}(t) = 0$$

for all roots inside the interval, then evaluate:

- $x(t_0)$;
- $x(t_f)$;
- $x(t_i)$ for every interior root t_i of $v(t)$.

The largest of these values is the maximum position, and the smallest is the minimum position.

5.6.2 Degree-9 Specialization

For a degree-9 polynomial:

$$x(t) = \sum_{i=0}^9 a_i t^i$$

the velocity is degree 8:

$$v(t) = \dot{x}(t)$$

So the position-extrema problem reduces to solving a degree-8 polynomial equation.

5.6.3 Important Interpretation

The second derivative test may help classify local extrema, but for global max/min over a finite interval it is not strictly necessary. The safest method is direct comparison of all candidate values.

5.7 Velocity Extrema In 1D

5.7.1 General Rule

To find maxima and minima of velocity $v(t)$ over $[t_0, t_f]$, solve:

$$a(t) = \ddot{x}(t) = 0$$

for all roots inside the interval, then evaluate:

- $v(t_0)$;
- $v(t_f)$;
- $v(t_i)$ for every interior root t_i of $a(t)$.

5.7.2 Degree-9 Specialization

For a degree-9 polynomial, acceleration is degree 7:

$$a(t) = \ddot{x}(t)$$

So the velocity-extrema problem reduces to solving a degree-7 polynomial equation.

5.7.3 Absolute Velocity

When a velocity limit is expressed as a magnitude bound:

$$|v(t)| \leq v_{max}$$

the relevant quantity is not the signed maximum alone, but:

$$\max_{t \in [t_0, t_f]} |v(t)|$$

This can be obtained by evaluating $|v(t)|$ at the same candidate points used for the signed extrema, or equivalently by checking both the maximum and minimum signed values.

5.8 Acceleration Extrema In 1D

5.8.1 General Rule

To find maxima and minima of acceleration $a(t)$, solve:

$$j(t) = x^{(3)}(t) = 0$$

inside the interval, then compare:

- $a(t_0)$;
- $a(t_f)$;
- $a(t_i)$ for each interior root t_i of $j(t)$.

5.8.2 Degree-9 Specialization

For the degree-9 polynomial, jerk is degree 6:

$$j(t) = x^{(3)}(t)$$

So acceleration-extrema analysis reduces to solving a degree-6 polynomial equation.

5.8.3 Absolute Acceleration

For an acceleration bound:

$$|a(t)| \leq a_{max}$$

the quantity to monitor is:

$$\max_{t \in [t_0, t_f]} |a(t)|$$

again obtained by evaluating the acceleration at all candidate points and taking absolute values.

5.9 Jerk Extrema In 1D

5.9.1 General Rule

To find maxima and minima of jerk $j(t)$, solve:

$$s(t) = x^{(4)}(t) = 0$$

inside the interval, then compare:

- $j(t_0)$;
- $j(t_f)$;
- $j(t_i)$ for each interior root t_i of $s(t)$.

5.9.2 Degree-9 Specialization

For the degree-9 polynomial, snap is degree 5:

$$s(t) = x^{(4)}(t)$$

So jerk-extrema analysis reduces to solving a degree-5 polynomial equation.

5.9.3 Absolute Jerk

For a jerk limit:

$$|j(t)| \leq j_{max}$$

the relevant quantity is:

$$\max_{t \in [t_0, t_f]} |j(t)|$$

computed from the same candidate-point set.

5.10 Snap Extrema In 1D

5.10.1 General Rule

To find maxima and minima of snap $s(t)$, solve:

$$x^{(5)}(t) = 0$$

inside the interval, then compare:

- $s(t_0)$;
- $s(t_f)$;
- $s(t_i)$ for each interior root t_i of $x^{(5)}(t)$.

5.10.2 Degree-9 Specialization

For the degree-9 polynomial, crackle is degree 4:

$$x^{(5)}(t)$$

So snap-extrema analysis reduces to solving a quartic equation. This is the highest-order extrema search in this document whose stationarity equation still falls inside the classical closed-form range.

5.11 Summary Of Stationarity Equations

The extrema-search structure for the degree-9 case can be summarized as:

Quantity analysed	Candidate interior points come from
position $x(t)$	roots of $v(t) = 0$
velocity $v(t)$	roots of $a(t) = 0$
acceleration $a(t)$	roots of $j(t) = 0$
jerk $j(t)$	roots of $s(t) = 0$
snap $s(t)$	roots of $x^{(5)}(t) = 0$

This table is operationally important: it tells which polynomial must be solved for each profile.

5.12 Symbolic Setup Vs Numerical Root Extraction

The project prefers symbolic treatment, but extrema detection introduces an unavoidable practical distinction:

- the polynomial expressions and the stationarity equations can be written symbolically;
- the actual extraction of all real roots in the interval may require numerical methods.

This happens because:

- quartic equations still admit closed-form symbolic solutions in principle;
- quintic and higher-degree polynomial equations do not, in general, admit a universal closed-form solution by radicals.

Therefore, for the degree-9 case:

- snap extrema can still be tied to a quartic stationarity equation;
- jerk, acceleration, velocity, and position extrema typically require numerical root extraction for the stationarity polynomial.

This is not a contradiction with the symbolic-first goal. The theory remains symbolic; the final root-finding step becomes numerical when algebra no longer admits a general closed form.

5.13 Generic 1D Extrema Algorithm

The following algorithm applies to any scalar quantity among:

$$x(t), \quad v(t), \quad a(t), \quad j(t), \quad s(t)$$

1. Select the target profile $f(t)$.
2. Compute its derivative $f'(t)$.
3. Solve $f'(t) = 0$ and retain only the real roots inside $[t_0, t_f]$.
4. Build the candidate set:

$$\{t_0, t_f, t_1, t_2, \dots, t_m\}$$

5. Evaluate $f(t)$ at every candidate point.
6. Compare those values to obtain:
 - global maximum;
 - global minimum;
 - if needed, maximum absolute value.

This is the baseline algorithm for Requirement 7.

5.14 Practical Root-Finding Note

For software implementation, the root-finding step should be understood as:

- formulate the stationarity polynomial exactly from the coefficients;
- compute all its roots numerically;
- discard complex roots;
- discard real roots lying outside the interval;
- keep the remaining roots as extrema candidates.

Possible numerical strategies include:

- companion-matrix eigenvalue extraction;
- robust polynomial root solvers;
- interval-based refinement after coarse root localization.

The choice of numerical solver belongs to implementation, not to the theory baseline. The theory only requires that all real interval roots be found reliably enough for extrema detection.

5.15 Imposing A Velocity Limit In 1D

5.15.1 Problem Form

The constraint is:

$$|v(t)| \leq v_{max} \quad \forall t \in [t_0, t_f]$$

The question is not merely whether the current segment satisfies the limit, but how to modify the formulation so that it does.

5.15.2 Verification Step

The first step is always verification:

1. determine the polynomial coefficients;
2. compute $\max |v(t)|$ over the interval using the extrema algorithm;
3. compare the result with v_{max} .

If:

$$\max |v(t)| \leq v_{max}$$

then the constraint is already satisfied.

5.15.3 Constraint-Imposition Strategy

If the limit is violated, the standard project-level remedy is to modify the segment duration and recompute the coefficients.

The reason is structural:

- the endpoint conditions remain the same;
- the polynomial degree remains the same;
- the time law changes because the duration changes;
- the coefficients are then recomputed from the updated duration.

So the practical velocity-limit procedure is:

1. choose a candidate duration T ;
2. determine the coefficients using `PolynomialCoefficientDetermination.md`;

3. compute $\max |v(t)|$;
4. if the bound is violated, increase T and repeat;
5. stop when the limit is satisfied.

5.15.4 Duration Search

A robust algorithmic form is:

1. start from an initial duration $T^{(0)}$;
2. build a feasibility test:

$$F_v(T) = \max_{t \in [t_0, t_f]} |v(t; T)| - v_{max}$$

3. if $F_v(T) \leq 0$, the duration is feasible;
4. otherwise enlarge T and test again;
5. once a feasible interval is bracketed, optionally refine the smallest acceptable duration by bisection or another one-dimensional search method.

This turns velocity-limit imposition into a duration-search problem.

5.16 Imposing An Acceleration Limit In 1D

5.16.1 Problem Form

The constraint is:

$$|a(t)| \leq a_{max} \quad \forall t \in [t_0, t_f]$$

5.16.2 Verification Step

1. determine the polynomial coefficients;
2. compute $\max |a(t)|$ using the extrema algorithm driven by $j(t) = 0$;
3. compare the result with a_{max} .

5.16.3 Constraint-Imposition Strategy

If the bound is violated, the same logic applies:

- modify the duration;
- recompute the polynomial;
- recompute the acceleration extrema;
- repeat until the bound is satisfied.

Define:

$$F_a(T) = \max_{t \in [t_0, t_f]} |a(t; T)| - a_{max}$$

and search for a duration such that:

$$F_a(T) \leq 0$$

The method is the same as for velocity, but the monitored profile is acceleration instead of velocity.

5.17 Imposing A Jerk Limit In 1D

5.17.1 Problem Form

The constraint is:

$$|j(t)| \leq j_{max} \quad \forall t \in [t_0, t_f]$$

5.17.2 Verification Step

1. determine the polynomial coefficients;
2. compute $\max |j(t)|$ using the extrema algorithm driven by $s(t) = 0$;
3. compare the result with j_{max} .

5.17.3 Constraint-Imposition Strategy

If the bound is violated, define:

$$F_j(T) = \max_{t \in [t_0, t_f]} |j(t; T)| - j_{max}$$

and search for a duration satisfying:

$$F_j(T) \leq 0$$

Again, the general procedure is:

1. modify duration;
2. recompute coefficients;
3. recompute jerk extrema;
4. iterate until feasible.

5.18 A Caution About Time Scaling

The previous sections intentionally describe limit imposition through recomputation, not through a universal closed-form scaling formula.

That caution matters because, in the general project setting:

- physical endpoint derivative values may be prescribed;
- the coefficient solution depends on the chosen duration;
- changing T therefore changes the coefficient problem itself.

So, in the most general case treated here, it is safest to:

- recompute the polynomial for each tested duration;
- then re-evaluate the constrained profile.

5.18.1 Special Normalized-Profile Case

There is, however, an important special case.

If the normalized profile shape is already fixed and one only rescales time, then derivative magnitudes scale as:

$$v \sim \frac{1}{T}, \quad a \sim \frac{1}{T^2}, \quad j \sim \frac{1}{T^3}$$

This explains why increasing duration often helps reduce peak derivatives.

But in the baseline formulation of this project, the more general recompute-and-verify strategy is the safer one to adopt for requirements 8, 9, and 10.

5.19 Position, Velocity, Acceleration, Jerk, And Snap In Normalized Time

Using normalized time can simplify extrema analysis because every segment is studied on the same interval:

$$\tau \in [0, 1]$$

The candidate-point logic remains identical:

- extrema of $x(\tau)$ come from $\frac{dx}{d\tau} = 0$;
- extrema of $\frac{dx}{d\tau}$ come from $\frac{d^2x}{d\tau^2} = 0$;
- and so on.

The benefit is mainly organizational:

- one local interval for all segments;
- one uniform search domain;
- easier comparison across segments.

The cost is that physical derivative limits must still be interpreted through the scaling relations between t and τ .

5.20 What Changes In 3D

The scalar extrema logic does not disappear in 3D. It reappears in two distinct ways:

- component-wise analysis;
- vector-magnitude analysis.

5.20.1 Component-Wise Analysis

Each Cartesian component is treated as its own scalar polynomial:

$$x(t), \quad y(t), \quad z(t)$$

Then each component has its own profiles:

- $v_x(t), v_y(t), v_z(t)$;
- $a_x(t), a_y(t), a_z(t)$;
- $j_x(t), j_y(t), j_z(t)$;
- $p_x^{(4)}(t), p_y^{(4)}(t), p_z^{(4)}(t)$.

The scalar extrema algorithm can then be applied independently to each component.

5.20.2 Vector-Magnitude Analysis

Alternatively, one may analyse magnitudes such as:

$$\|\mathbf{v}(t)\|, \quad \|\mathbf{a}(t)\|, \quad \|\mathbf{j}(t)\|, \quad \|\mathbf{p}^{(4)}(t)\|$$

This is often more physically meaningful when the concern is the total kinematic demand rather than the per-axis demand.

For example:

$$\|\mathbf{v}(t)\| = \sqrt{v_x(t)^2 + v_y(t)^2 + v_z(t)^2}$$

To find extrema of velocity magnitude, one can instead study:

$$\|\mathbf{v}(t)\|^2 = v_x(t)^2 + v_y(t)^2 + v_z(t)^2$$

because squaring preserves the location of maxima and minima for nonnegative magnitudes.

Its stationarity condition is:

$$\frac{d}{dt} (\|\mathbf{v}(t)\|^2) = 2 \mathbf{v}(t) \cdot \mathbf{a}(t) = 0$$

Similarly:

- extrema of $\|\mathbf{a}(t)\|^2$ come from $\mathbf{a}(t) \cdot \mathbf{j}(t) = 0$;
- extrema of $\|\mathbf{j}(t)\|^2$ come from $\mathbf{j}(t) \cdot \mathbf{p}^{(4)}(t) = 0$.

5.21 Recommended 3D Interpretation For This Project

This document resolves the earlier open question deferred from `SymbolsAndNomenclature.md` and `NinthDegreePolynomialTheory.md`.

The recommended project baseline is:

- use **component-wise analysis as the primary theory and implementation baseline**;
- use **vector-magnitude analysis as an additional derived check when the total 3D kinematic demand matters**.

This recommendation is consistent with the project assumptions because:

- the project is explicitly built from 1D to 3D;
- component-wise treatment is the direct extension of the scalar theory;
- it is simpler to derive, verify, and implement;
- it avoids introducing unnecessary coupling too early.

So, for the continuation of this project:

- the default interpretation of 3D constraints is component-wise;
- vector-magnitude analysis remains valid and useful, but secondary.

5.22 Relation To The Data Model

`02-data-model.md` defines:

$$\text{Constraints} = \{\text{constraintType}, \text{constraintValue}, \text{constantVelocitySelection}, \dots\}$$

From the viewpoint of this document:

- **constraintType** can represent velocity, acceleration, jerk, snap, time, or related segment-level restrictions;
- **constraintValue** is the bound associated with the selected type;
- **constraints[]** inside each segment is the place where one or more such restrictions are attached to that segment.

This document gives the theoretical meaning of those entries:

- analysis of maxima/minima tells whether a given profile violates a bound;

- duration search provides one baseline strategy to impose the bound;
- the same structure later supports the constant-velocity requirement addressed separately in `ConstantVelocitySegment`.

One implementation-oriented point remains open: if multiple limits must be imposed simultaneously on the same segment, the future software design may need a clearer representation than a single type/value pair viewed in isolation, even though the current `constraints[]` container already points in that direction.

5.23 Generic Constraint-Verification Algorithm

For a selected derivative order and a selected bound, the following procedure can be used.

1. Determine the segment coefficients.
2. Build the target profile:
 - $v(t)$ for velocity;
 - $a(t)$ for acceleration;
 - $j(t)$ for jerk.
3. Compute the candidate extrema points from the next derivative.
4. Evaluate the target profile at all candidates.
5. Compute the maximum absolute value.
6. Compare that result with the admissible limit.
7. Mark the segment as feasible or infeasible.

This algorithm is the verification core shared by requirements 8, 9, and 10.

5.24 Generic Constraint-Imposition Algorithm

The baseline limit-imposition strategy for the current project is:

1. choose the segment degree and boundary conditions;
2. choose an initial duration;
3. determine the polynomial coefficients;
4. verify the selected constraint;
5. if violated, increase duration and recompute;
6. repeat until the constraint is satisfied;
7. if desired, refine the smallest feasible duration.

This is conceptually simple, consistent with the current theory baseline, and directly compatible with the future desktop application.

5.25 Worked Symbolic Considerations

5.25.1 Example 1: Position Extrema

Suppose a degree-9 segment is written as:

$$x(t) = \sum_{i=0}^9 a_i t^i$$

To determine the maximum and minimum position:

1. compute $v(t)$;
2. solve $v(t) = 0$ in $[t_0, t_f]$;
3. evaluate $x(t)$ at the interval endpoints and at the retained interior roots;
4. compare the values.

The result is exact at the level of the candidate-point principle, even if the interior roots themselves are obtained numerically.

5.25.2 Example 2: Velocity Limit

Suppose the segment must satisfy:

$$|v(t)| \leq v_{max}$$

The procedure is:

1. compute the coefficients for the current duration;
2. solve $a(t) = 0$ inside the interval;
3. evaluate $|v(t)|$ at all candidate points;
4. if the largest value exceeds v_{max} , enlarge duration and recompute.

This gives a clean theoretical basis for the future implementation of a velocity-limit tool.

5.25.3 Example 3: 3D Constraint Interpretation

Suppose a 3D trajectory has velocity vector:

$$\mathbf{v}(t) = \begin{bmatrix} v_x(t) \\ v_y(t) \\ v_z(t) \end{bmatrix}$$

The project baseline first checks:

$$|v_x(t)|, \quad |v_y(t)|, \quad |v_z(t)|$$

component by component.

If needed, a secondary check may also inspect:

$$\|\mathbf{v}(t)\|$$

to understand the total 3D kinematic demand.

5.26 Open Questions

- **Open — to resolve during software implementation (src/):** which numerical root solver should be adopted for interval root extraction of the stationarity polynomials?
- **Open — to resolve during software design:** when several limits are imposed simultaneously on the same segment, should the application search duration only, or also allow reformulations that change additional parameters?
- **Open — to resolve during later project phases:** should snap also receive an explicit limit-imposition requirement, beyond the extrema analysis already covered here?

5.27 Related Documents

- 06-theory-library-roadmap.md
- SymbolsAndNomenclature.md
- NinthDegreePolynomialTheory.md
- BoundaryConditions.md
- PolynomialCoefficientDetermination.md
- ConstantVelocitySegment
- 02-data-model.md

6 Constant Velocity Segment

6.1 Intro And Scope

This document studies how to introduce a constant-velocity segment inside a trajectory originally described through a ninth-degree polynomial.

It covers Requirement 12 of `01-requirements.md`:

- a constant-speed segment is required inside a path originally defined as a degree-9 polynomial;
- the segment is specified by its start and end positions, either in absolute form or as percentages;
- the treatment starts in 1D and then extends to 3D.

This document builds on:

- `PolynomialCoefficientDetermination.md`, for segment recomputation from boundary data;
- `TrajectoryConstraints.md`, for the interpretation of velocity limits and duration search;
- `02-data-model.md`, for the `Constraints` fields that already reserve `constantVelocitySelection`, `constantVelocityValue`, and the start/end positions of the constant-velocity part.

The discussion remains symbolic-first and theoretical, consistent with `03-assumptions.md`.

6.2 Symbols Used

The notation follows `SymbolsAndNomenclature.md`. The main symbols used here are:

- t : absolute time;
- t_0 : initial time;
- t_f : final time;
- $T = t_f - t_0$: duration of a generic segment;
- $\tau = \frac{t-t_0}{T}$: normalized time;
- $x(t)$: scalar position law in 1D;
- $\mathbf{p}(t)$: position vector in 3D;
- $v(t) = \dot{x}(t)$: scalar velocity in 1D;
- $\mathbf{v}(t) = \dot{\mathbf{p}}(t)$: velocity vector in 3D;
- v_{cv} : requested constant speed magnitude inside the constant-velocity segment;
- $x_{cv}^{start}, x_{cv}^{end}$: start and end positions of the constant-velocity subsegment in 1D;
- $\mathbf{p}_{cv}^{start}, \mathbf{p}_{cv}^{end}$: start and end positions of the constant-velocity subsegment in 3D;
- $t_{cv}^{start}, t_{cv}^{end}$: start and end times of the constant-velocity subsegment;
- $T_{cv} = t_{cv}^{end} - t_{cv}^{start}$: duration of the constant-velocity subsegment.

When the constant-velocity segment is embedded between transition segments, the endpoint states of that middle segment are written as:

$$\mathcal{S}_{cv}^{start}, \quad \mathcal{S}_{cv}^{end}$$

when compact notation is useful.

6.3 Path Vs Trajectory: Why This Topic Is Special

This requirement is easier to understand if the distinction from `04-future-topics.md` is made explicit:

- a **path** is the geometric locus to be followed;
- a **trajectory** is that path plus a time law.

A constant-velocity segment changes the time law in a very specific way:

- in 1D, it imposes constant signed velocity or constant speed magnitude over a selected interval;

- in 3D, it raises a choice between constant component-wise velocity and constant speed along the geometric path.

So the central question is not only “how do we keep velocity constant?”, but also:

- what exactly must remain unchanged from the original polynomial path;
- what is allowed to be reformulated;
- what continuity level is expected at the boundaries of the constant-velocity part.

6.4 First Key Fact: Exact Constant Velocity Is Very Restrictive

Suppose a single polynomial segment contains a nontrivial interval on which velocity is exactly constant.

In 1D, that means:

$$\dot{x}(t) = v_{cv}$$

for every t in some interval of nonzero length.

Equivalently:

$$\ddot{x}(t) = 0$$

throughout that interval.

But $\ddot{x}(t)$ is itself a polynomial. A polynomial that is zero on a nontrivial interval is zero everywhere. Therefore:

$$\ddot{x}(t) \equiv 0$$

for the whole segment, and the segment must be globally affine:

$$x(t) = c_0 + c_1 t$$

So, in general:

- an arbitrary ninth-degree polynomial cannot contain an exact constant-velocity subinterval while remaining the same single ninth-degree polynomial;
- an exact constant-velocity part therefore requires a **reformulation** of the trajectory into multiple segments, unless the whole original segment is already linear.

This fact is foundational for the rest of the document.

6.5 1D Case: Exact Constant Velocity Segment

6.6 Geometric Interpretation

In 1D, the path is simply a line on the scalar axis. So “constant speed along the path” and “constant velocity magnitude” are almost the same notion, apart from sign.

If the motion direction over the selected interval is fixed, then the exact constant-velocity segment is:

$$x_{cv}(t) = x_{cv}^{start} + \sigma v_{cv}(t - t_{cv}^{start})$$

where:

- $\sigma \in \{+1, -1\}$ encodes the travel direction;
- $v_{cv} > 0$ is the requested speed magnitude.

The corresponding derivatives are:

$$\dot{x}_{cv}(t) = \sigma v_{cv}, \quad \ddot{x}_{cv}(t) = 0, \quad x_{cv}^{(3)}(t) = 0, \quad x_{cv}^{(4)}(t) = 0$$

for the whole constant-velocity interval.

6.7 Duration Of The Constant-Velocity Part

Once the selected positions are known, the duration of the middle constant-velocity part is:

$$T_{cv} = \frac{|x_{cv}^{end} - x_{cv}^{start}|}{v_{cv}}$$

provided:

$$x_{cv}^{end} \neq x_{cv}^{start}, \quad v_{cv} > 0$$

If the two positions coincide, the constant-velocity subsegment degenerates and the requirement loses meaning.

6.8 Why A Multi-Segment Reformulation Is Needed

If the original trajectory outside the selected interval is a degree-9 polynomial, then an exact constant-velocity part is normally introduced as a three-part structure:

1. entry transition segment;
2. constant-velocity middle segment;
3. exit transition segment.

In symbolic form:

$$\mathcal{T} = \{\text{entry transition, constant-velocity segment, exit transition}\}$$

The role of the two transition segments is to connect the surrounding motion to a middle segment whose higher derivatives are all zero.

6.9 Entry And Exit States In 1D

The middle segment has the natural endpoint states:

$$\mathcal{S}_{cv}^{start} = (x_{cv}^{start}, \sigma v_{cv}, 0, 0, 0)$$

$$\mathcal{S}_{cv}^{end} = (x_{cv}^{end}, \sigma v_{cv}, 0, 0, 0)$$

These states make the nature of the middle segment explicit:

- prescribed position at both ends;
- same velocity at both ends;
- zero acceleration, jerk, and snap at both ends.

The entry transition must end at $\mathcal{S}_{cv}^{start}$.

The exit transition must begin at $\mathcal{S}_{cv}^{end}$.

6.10 Baseline 1D Construction Strategy

The cleanest exact strategy in the current project is:

1. choose the two positions that delimit the constant-velocity interval;
2. define the target speed magnitude v_{cv} ;
3. compute the constant-velocity duration T_{cv} ;
4. create a middle affine segment with velocity σv_{cv} ;
5. construct an entry degree-9 transition that reaches $\mathcal{S}_{cv}^{start}$;
6. construct an exit degree-9 transition that leaves $\mathcal{S}_{cv}^{end}$ toward the later trajectory conditions;
7. determine the polynomial coefficients of the transition segments using `PolynomialCoefficientDetermination.md`.

This is the main exact 1D strategy recommended by this document.

6.11 How The Start And End Positions May Be Specified In 1D

The requirement allows the interval to be specified either in absolute form or by percentages.

6.11.1 Absolute Specification

The user gives:

$$x_{cv}^{start}, \quad x_{cv}^{end}$$

directly.

This is the clearest formulation whenever the scalar coordinate range is already meaningful.

6.11.2 Percentage Specification

A simple percentage-based choice is:

$$\lambda_{start}, \lambda_{end} \in [0, 1]$$

with:

$$x_{cv}^{start} = x_0 + \lambda_{start}(x_f - x_0)$$

$$x_{cv}^{end} = x_0 + \lambda_{end}(x_f - x_0)$$

This is convenient, but it should be interpreted carefully:

- it is natural when the 1D motion is monotonic;
- it becomes less meaningful if the original polynomial overshoots or reverses direction.

So, in 1D, percentage specification is best treated as a convenience rule for monotonic path portions.

6.12 1D Consistency Checks

Before building the constant-velocity part, the following conditions should be checked.

1. The selected positions must lie on the intended path portion.
2. The ordering of the positions must be compatible with the chosen travel direction.
3. The requested speed must be positive:

$$v_{cv} > 0$$

4. The surrounding transition segments must have enough room to connect smoothly to zero higher derivatives at the boundaries of the constant-velocity segment.

The fourth point is especially important: a constant-velocity middle segment is easy to define, but smooth entry and exit may force the surrounding intervals to be long enough.

6.13 A Simpler But Less Smooth 1D Alternative

There is also a simpler formulation:

- keep only one polynomial segment before the constant part;
- insert the affine constant-velocity part;
- switch directly to the following part.

This is easy to define, but generally poor in smoothness because direct switching may produce discontinuities in acceleration, jerk, or snap.

So, unless low continuity is explicitly acceptable, the project baseline should prefer transition segments that match zero acceleration, jerk, and snap at the entry and exit of the constant-velocity part.

6.14 3D Case: Two Different Meanings Of “Constant Velocity”

In 3D, the topic becomes richer because two meanings must be separated.

6.14.1 Constant Velocity Vector

One may require:

$$\mathbf{v}(t) = \mathbf{v}_{cv}$$

constant over the selected interval.

Then the position law is:

$$\mathbf{p}_{cv}(t) = \mathbf{p}_{cv}^{start} + \mathbf{v}_{cv}(t - t_{cv}^{start})$$

This implies:

- straight-line motion along the chord joining the two selected points;
- constant speed magnitude;
- zero acceleration, jerk, and snap.

But this does **not** generally preserve the original curved polynomial path between the two selected positions.

6.14.2 Constant Speed Along The Original 3D Path

Alternatively, one may require that the motion follows the same geometric curve portion while travelling with constant speed magnitude:

$$\|\mathbf{v}(t)\| = v_{cv}$$

This preserves the path geometry, but the component velocities are no longer constant in general.

These two meanings coincide only in the special case where the selected path portion is already a straight line.

6.15 Decided 3D Interpretation For This Requirement

DECISION: for this requirement, the project adopts arc-length reparameterization (path-preserving, constant speed magnitude) as the baseline 3D interpretation, not just as a recommendation.

- preserve the selected geometric subpath of the original polynomial curve;
- impose constant **speed magnitude** along that preserved subpath.

Rationale:

- this interpretation offers the widest applicative range: it correctly handles a constant-speed requirement along a genuinely curved, multi-axis path, which the straight-chord alternative cannot do without deviating from the original path;
- in the practical case where the constant-speed requirement applies to motion along a single Cartesian axis — the most common real-world case for this kind of process constraint (e.g. avoiding acceleration-induced stress on the transported/machined body along one direction) — the original path in that interval is already a straight line, and the general arc-length construction reduces exactly (not approximately) to the same affine law used in the 1D case. No separate code path is needed for that case;
- the straight-chord alternative remains valid and simpler, but only coincides with path preservation when the selected subpath is already straight; for a genuinely curved multi-axis subpath it silently departs from the original geometry;
- the straight-chord-on-a-genuinely-curved-multi-axis-path case is deliberately not adopted as the project baseline now — it is deferred and will be added only if a concrete need for it arises (see Open Questions).

This decision was reached after clarifying two points: (a) only the straight-chord option keeps the velocity *vector* constant in both magnitude and direction — the arc-length option keeps only the magnitude constant, since direction varies continuously along a curve; (b) the two options coincide exactly, not just approximately, whenever the selected subpath is a straight line — which was verified algebraically by reducing the general arc-length construction to the 1D affine formula for that special case.

So, unlike the component-wise baseline adopted in `TrajectoryConstraints.md` for general constraint analysis, this document is primarily path-oriented:

- for general limits, component-wise treatment remains the default baseline;
- for this specific requirement, the decided baseline is constant speed along the selected path portion, via arc-length reparameterization.

6.16 3D Path-Preserving Strategy: Arc-Length Reparameterization

Let the original 3D polynomial path be:

$$\mathbf{p}_{orig}(t)$$

and let the selected subpath correspond to original parameter values:

$$t_s^{orig}, \quad t_e^{orig}$$

with:

$$\mathbf{p}_{cv}^{start} = \mathbf{p}_{orig}(t_s^{orig}), \quad \mathbf{p}_{cv}^{end} = \mathbf{p}_{orig}(t_e^{orig})$$

Define the arc-length function from the start of the selected interval:

$$\ell(t) = \int_{t_s^{orig}}^t \|\dot{\mathbf{p}}_{orig}(u)\| \, du$$

Then the total length of the selected subpath is:

$$L_{cv} = \ell(t_e^{orig})$$

If the requested constant speed magnitude is v_{cv} , the duration of the constant-speed traversal becomes:

$$T_{cv} = \frac{L_{cv}}{v_{cv}}$$

Now define a new local time variable over the constant-speed part:

$$\sigma = \frac{t - t_{cv}^{start}}{T_{cv}} \in [0, 1]$$

and the corresponding traveled arc length:

$$s(\sigma) = \sigma L_{cv}$$

The path-preserving constant-speed motion is then obtained by:

$$\mathbf{p}_{cv}(t) = \mathbf{p}_{orig}(\ell^{-1}(s(\sigma)))$$

This construction preserves the original curve and enforces constant speed magnitude along it.

6.17 Important Consequence In 3D

The arc-length-reparameterized constant-speed segment is generally **not polynomial in time anymore**.

This matters because it means:

- the original geometric path can be preserved;
- exact constant speed can be preserved;
- but the resulting time law is no longer, in general, a degree-9 polynomial in the new time variable.

So the 3D problem naturally separates into two layers:

- geometric path preservation;
- time-law reformulation.

This is not a defect. It is the mathematically natural outcome of the requirement.

6.17.1 Implementation Note: Where Numerical Methods Are Actually Needed

This non-polynomial consequence only bites in the genuinely curved, multi-axis case:

- **single-axis case** (the original subpath is already a straight line): as shown above, the arc-length construction reduces exactly to the closed-form affine law — evaluation stays polynomial, no numerical methods needed;
- **genuinely curved multi-axis case**: $\ell(t)$ generally has no closed form for a degree-9 polynomial path and must be computed via **numerical integration**; recovering $\ell^{-1}(s)$ then requires **numerical root-finding**. This is the only point in the theory library so far where evaluation departs from direct coefficient/Horner-based computation.

Implementation should branch on this distinction rather than always paying the numerical cost: detect the single-axis (straight-subpath) case and use the closed-form affine law directly, reserving numerical integration and root-finding for the genuinely curved multi-axis case.

6.18 3D Straight-Chord Alternative

If path preservation is not required, then the simpler exact construction is a straight-line constant-velocity segment:

$$\mathbf{p}_{cv}(t) = \mathbf{p}_{cv}^{start} + \frac{t - t_{cv}^{start}}{T_{cv}} (\mathbf{p}_{cv}^{end} - \mathbf{p}_{cv}^{start})$$

with:

$$T_{cv} = \frac{\|\mathbf{p}_{cv}^{end} - \mathbf{p}_{cv}^{start}\|}{v_{cv}}$$

This gives:

- constant velocity vector;
- constant speed magnitude;
- straight motion between the selected points.

Its advantage is simplicity.

Its limitation is that the curved portion of the original polynomial path is replaced by its chord.

6.19 How The Start And End Positions May Be Specified In 3D

6.19.1 Absolute Specification

The user gives:

$$\mathbf{p}_{cv}^{start}, \quad \mathbf{p}_{cv}^{end}$$

directly.

This is simple only if those points are already known on the original path.

6.19.2 Percentage Specification Along The Path

For path-oriented use, the most meaningful percentage convention is based on the path progression parameter over the selected original interval.

One convenient choice is to define:

$$\lambda_{start}, \lambda_{end} \in [0, 1]$$

on the original segment parameter interval:

$$t_s^{orig} = t_0 + \lambda_{start}(t_f - t_0)$$

$$t_e^{orig} = t_0 + \lambda_{end}(t_f - t_0)$$

and then:

$$\mathbf{p}_{cv}^{start} = \mathbf{p}_{orig}(t_s^{orig}), \quad \mathbf{p}_{cv}^{end} = \mathbf{p}_{orig}(t_e^{orig})$$

This is parameter-percentage, not true arc-length percentage.

If a geometric percentage of traveled distance is preferred, then the percentage should be defined on arc length instead:

$$\ell(t_s^{orig}) = \lambda_{start} L_{tot}, \quad \ell(t_e^{orig}) = \lambda_{end} L_{tot}$$

where L_{tot} is the total arc length of the original segment.

For this project, the cleanest recommendation is:

- in 1D, percentage by scalar position interval is acceptable on monotonic portions;
- in 3D, percentage by original parameter is the simplest practical convention;
- if precise geometric meaning is important in 3D, arc-length percentage is the more rigorous convention.

6.20 Transition Segments Around The Constant-Speed Part

Whether the constant-speed segment is 1D or 3D, smooth entry and exit normally require dedicated transition segments.

6.20.1 1D Transition Targets

In 1D, the target state at the start of the constant-speed interval is:

$$(x_{cv}^{start}, \sigma v_{cv}, 0, 0, 0)$$

and at the end:

$$(x_{cv}^{end}, \sigma v_{cv}, 0, 0, 0)$$

6.20.2 3D Transition Targets

For the straight-chord constant-velocity-vector option:

$$\mathbf{v}_{cv} = \frac{\mathbf{p}_{cv}^{end} - \mathbf{p}_{cv}^{start}}{T_{cv}}$$

and the transitions should match:

- position;
- velocity vector $\mathbf{v}_{cv}$;
- zero acceleration;
- zero jerk;
- zero snap.

For the arc-length-reparameterized path-preserving option, the velocity direction varies along the segment, so transition design is less trivial. In that case, the main requirement is smooth matching between:

- the preceding motion state;
- the reparameterized curve state at the beginning of the constant-speed interval;
- the corresponding state at the end.

This may lead to dedicated blending/reparameterization design choices in later documents.

6.21 Baseline Algorithm For Requirement 12

The following algorithm is the recommended baseline for the project.

1. Start from the original degree-9 trajectory segment.
2. Select the start and end of the desired constant-speed interval:
 - in absolute coordinates, or
 - through a chosen percentage convention.
3. Determine the corresponding positions:
 - $x_{cv}^{start}, x_{cv}^{end}$ in 1D, or
 - $\mathbf{p}_{cv}^{start}, \mathbf{p}_{cv}^{end}$ in 3D.
4. Choose the requested constant speed magnitude v_{cv} .
5. Build the middle constant-speed segment:
 - affine in 1D;
 - arc-length-reparameterized in 3D if path preservation is required;
 - straight-chord affine in 3D if path preservation is not required.
6. Compute the duration T_{cv} of the middle segment.
7. Construct entry and exit transitions that connect smoothly to the middle segment.
8. Compute the transition coefficients with `PolynomialCoefficientDetermination.md`.
9. Verify continuity and duration consistency for the full three-part construction.

6.22 Relation To The Data Model

`02-data-model.md` already defines the following fields inside `Constraints`:

- `constantVelocitySelection`
- `constantVelocityValue`
- `constantVelocityInitialPos1D`
- `constantVelocityEndPos1D`
- `constantVelocityInitialPos`
- `constantVelocityEndPos`

This document provides the theoretical meaning of those fields:

- `constantVelocitySelection` activates the special subsegment logic;
- `constantVelocityValue` stores the requested speed magnitude v_{cv} ;
- the 1D and 3D position fields locate the start and end of the constant-speed part.

What the data model does **not** yet decide explicitly is:

- which percentage convention should be used when positions are not given absolutely;
- how transition segments are represented as part of the higher-level trajectory structure.

(Whether the 3D interpretation is path-preserving or straight-chord is no longer open — see *Decided 3D Interpretation For This Requirement* above.)

Those are theory-to-design bridges, not contradictions.

6.23 Worked Symbolic Considerations

6.23.1 Example 1: 1D Exact Constant-Velocity Middle Segment

Suppose the selected interval is:

$$x_{cv}^{start} \rightarrow x_{cv}^{end}$$

with requested speed magnitude v_{cv} .

Then the middle segment is:

$$x_{cv}(t) = x_{cv}^{start} + \sigma v_{cv}(t - t_{cv}^{start})$$

with duration:

$$T_{cv} = \frac{|x_{cv}^{end} - x_{cv}^{start}|}{v_{cv}}$$

The surrounding degree-9 transition segments must reach and leave this affine law with zero acceleration, jerk, and snap at the junctions.

6.23.2 Example 2: 3D Path-Preserving Constant-Speed Segment

Suppose the original polynomial path is:

$$\mathbf{p}_{orig}(t)$$

and two points on it are selected.

The geometric subpath between them is preserved, but the time law is replaced by an arc-length-based one:

$$\mathbf{p}_{cv}(t) = \mathbf{p}_{orig}(\ell^{-1}(s(\sigma)))$$

This preserves the original curve while enforcing constant speed magnitude.

6.23.3 Example 3: 3D Straight-Chord Alternative

If one instead chooses:

$$\mathbf{p}_{cv}(t) = \mathbf{p}_{cv}^{start} + \frac{t - t_{cv}^{start}}{T_{cv}} (\mathbf{p}_{cv}^{end} - \mathbf{p}_{cv}^{start})$$

then the motion is exactly constant-velocity in the vector sense, but the original curved subpath is replaced by a straight chord.

This makes the trade-off explicit:

- preserve path geometry, or
- preserve a simpler affine time law.

6.24 Open Questions

- **Open — to resolve if/when a concrete need arises:** the straight-chord alternative for a genuinely curved multi-axis subpath is deliberately not part of the current baseline (see *Decided 3D Interpretation*); should it be added later as an explicit user-facing option, and what would trigger that need?
- **Open — to resolve during data-format design:** when users specify the interval in percentages, which convention should be standardized in 3D: parameter percentage, arc-length percentage, or both?
- **Open — to resolve together with the later blending documents:** should the transition segments around the constant-speed part always enforce zero acceleration, jerk, and snap at the junctions, or should the continuity order be configurable?

6.25 Related Documents

- `06-theory-library-roadmap.md`
- `SymbolsAndNomenclature.md`
- `PolynomialCoefficientDetermination.md`
- `TrajectoryConstraints.md`
- `BlendingSegments`
- `TrajectoryPassingThroughConstraintPoints`
- `02-data-model.md`